

Global Form and Generalized Symmetry of the Standard Model

Central quotients, genuine line operators, and defect-sector normal forms

SYM-9 of the Standard Model Symmetry Arc

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Principal result

Fix the Standard Model local gauge algebra

$$\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_{\tilde{Y}},$$

the anomaly-neutral chiral matter and one-Higgs-doublet representation ledger, integer hypercharge convention, and a declared operator category. The common center element

$$\xi = (\omega_3, -1, e^{i\pi/3}), \quad \omega_3 = e^{2\pi i/3},$$

acts trivially on the inherited local matter package and generates $K_{\text{mat}} \cong \mathbb{Z}_6$. The compatible compact connected global-form targets are therefore

$$G_{\Gamma} = \frac{\text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_{\tilde{Y}}}{\Gamma}, \quad \Gamma \in \{1, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_6\}.$$

At fixed operator, magnetic, line-polarization, background-action, and phase scope, each quotient constrains electric descent, magnetic and dyonic charges, screening, one-form actions, and gauge-topological theta identifications. The gauge group and DSZ pairing do not by themselves select a unique maximal mutually local dyonic branch, and the gauge-topological torus is not the full Standard Model phase space until anomalous fermion rephasings and mass/Yukawa phases are included. Higher-flavor 2-group and noninvertible defect structures occur only on target fibers carrying their required assumptions. These data assemble into a global-form-indexed packet invariant under lawful presentation changes; changing genuine extended, background-response, or physical phase content changes the completed target.

Document status	Publication edition v1.3.0
Proof order	Framework-neutral QFT analysis; quotient, line, and topological reconstruction; AASC role assignment; scope-preserving rival analysis
Direct SYM imports	SYM-0, SYM-1, SYM-3, SYM-5, SYM-7, and the electroweak handoff from SYM-8
Primary QFT interfaces	Global form and line operators; generalized symmetry; higher groups; Standard Model higher flavor; noninvertible defects
Successor boundary	Scale/RG transport to SYM-10; candidate completeness and whole-arc exhaustion to SYM-11

Abstract

The local Standard Model gauge algebra and its perturbative particle representations do not specify the global gauge theory. This paper fixes the anomaly-neutral chiral matter and one-Higgs-doublet ledger in integer hypercharge normalization, computes the common central kernel $K_{\text{mat}} \cong \mathbb{Z}_6$, and classifies the compatible compact connected global-form targets G_Γ for $\Gamma = 1, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_6$. For each target with a fixed maximal mutually local line-polarization branch, it reconstructs the electric descent conditions, the magnetic and dyonic charge lattice, the screening quotient, and the resulting one-form symmetry. The electric one-form factor is $\mathbb{Z}_6/\Gamma$, while the no-dynamical-monopole target carries a continuous magnetic $U(1)^{(1)}$ whose normalization and breaking depend on the admitted magnetic sector.

The global form also changes the coupled identification lattice of the non-Abelian and hypercharge theta coordinates. The manuscript distinguishes the resulting gauge-topological line torus

$$\Theta_\Gamma^{\text{line}} = \mathbb{R}^3 / \Lambda_{\theta, \Gamma; \mathcal{P}_\Gamma}$$

from the full physical phase packet, which additionally quotients the mass/Yukawa phase data by admitted anomalous fermion rephasings. This separation retains the Witten-effect line information without mistaking a fixed matter-phase convention for the complete Standard Model phase space. Distinct global completions can agree on all local perturbative correlators yet differ in line polarization, genuine line operators, monopoles, background-action or fractionalization data, topological response, theta identifications, and physical phase classes. They are distinct completed targets whenever the operator category includes those observables.

The paper then types two frontier structures without importing them into the generic massive core. In the zero-Yukawa background-probe target, flavor currents and the magnetic hypercharge one-form current can form a continuous 2-group with quantized triangle coefficient; the numerical Postnikov coordinate is transported with the magnetic-current and two-form-background normalization. More generally, when a zero-form background is admitted together with a discrete one-form factor, the action on charged lines may require a separate fractionalization class. In massless QED and first-generation massless QCD below the electroweak scale, rational axial rotations can be completed by topological field theories into noninvertible symmetry defects whose action is noninvertible on 't Hooft lines. These are target-fiber results, not universal symmetries of the observed massive Standard Model.

Only after the framework-neutral field-theory analysis is complete is the AASC role map applied. Global form is a target-bearing structural datum; genuine generalized actions are physical on extended operators; line, monopole, theta, background, and defect coordinates remain representatives or witnesses according to locus. The integrated endpoint is a global-form-indexed defect-sector normal form and a precise handoff to SYM-11: symmetry closure must be stated at a fixed Γ and complete operator/background/phase target, or explicitly over the family of completed targets.

Keywords: Standard Model gauge group; global form; central quotient; Wilson line; 't Hooft line; dyonic line; one-form symmetry; theta periodicity; physical phase quotient; 2-group symmetry; noninvertible symmetry; AASC.

Proof and certification status

The scientific conclusions are established by the mathematical and physical argument in the manuscript, using the cited primary literature where results are imported. The accompanying archive audits target scope, source authority, theorem identity, encoded center and lattice arithmetic, table synchronization, bounded rival records, artifact identity, and reproducibility. It does not substitute for the field-theory proof, select a unique physical value of Γ , or establish generalized-symmetry exhaustion beyond the declared targets.

Reader's guide: target family, proof spine, and claim hierarchy

SYM-9 is the global-completion stage of the Standard Model Symmetry Arc. SYM-1 established that a local gauge algebra, connection, curvature, and proper local gauge quotient do not settle the global gauge group, boundary charges, or line-operator category. SYM-3 separated perturbative gauge-anomaly cancellation from global and generalized anomaly questions. SYM-5 fixed the distinction between flavor actions, basis changes, spurions, and invariant data. SYM-7 fixed the operator-dimension and neutrino-target dependence of accidental symmetries. SYM-8 reconstructed the one-doublet electroweak fixation, residual electromagnetic generator, physical charge ledger, and perturbative spectrum while explicitly deferring faithful global form, line operators, and monopoles. The present manuscript takes that deferred step.

The argument obeys the arc-wide order

framework-neutral global-gauge and extended-operator analysis

→ quotient, line, screening, and topological certificates

→ AASC role assignment → scope-preserving rival analysis.

No AASC label is used to prove the preceding field-theory statement. In particular, “target datum,” “physical action,” “witness,” and “skin” do not determine the common center, the descent of a Wilson line, the Dirac pairing, the one-form group, a theta identification, a symmetry-fractionalization class, a 2-group structure constant, or a noninvertible fusion rule. Those are established first.

Scientific proof spine

1. Freeze the local matter target, integer hypercharge convention, spin assumptions, operator category, magnetic-sector policy, line-spectrum branch, background transformation package, fractionalization/extension class, flavor/Yukawa status, and defect completion.
2. Prove local-to-global underdetermination: the Lie algebra and local perturbative observables do not determine the faithful global group or the spectrum of genuine extended operators.
3. Compute the common center action on every Standard Model multiplet and derive $K_{\text{mat}} = \mathbb{Z}_6$.
4. Classify the four subgroup quotients G_{Γ} compatible with the local representation ledger, while distinguishing the quotient acting faithfully on local matter from the smaller-quotient theories that retain a nontrivial electric one-form symmetry.
5. Derive the electric descent congruences for Wilson labels, then separate global descent from dynamical screening.
6. Impose the Dirac–Schwinger–Zwanziger mutual-locality condition and reconstruct the magnetic and dyonic line lattices on the fixed maximal mutually local line-spectrum branch for each global form.
7. Derive the one-form symmetry from the unscreened line-charge structure and the magnetic Bianchi current, with explicit dependence on dynamical monopoles.
8. Construct the coupled gauge-topological theta-identification lattices, form the anomalous-rephasing quotient with the mass/Yukawa phase data, and transport the selected line branch and monopole data through one-doublet electroweak fixation.
9. Prove the global-form separation theorem: different admitted line, background-action, or topological packets define different completed targets even when their local perturbative sectors coincide.
10. State the higher-flavor 2-group and noninvertible defect results only on the target fibers carrying their required assumptions.
11. Apply the AASC role map and assemble the global-completion packet and family-indexed SYM-11 handoff.

Claim hierarchy

The manuscript distinguishes four kinds of statement.

1. **Local-coordinate statements:** a chosen representative for the center generator, hypercharge unit, electric/magnetic charge vector, theta coordinate, background gauge field, fractionalization cocycle, or defect diagram.
2. **Global QFT statements:** quotient group, descent condition, selected maximal mutually local branch, screening class, one-form symmetry, topological periodicity, background-action law, higher-group law, or defect fusion/action at a fixed target.
3. **Structural-role statements:** whether the item is target-bearing architecture, a physical action, a witness, a representative, an effective route, an obstruction, or a target update.
4. **Artifact-audit statements:** whether the published tables, encoded arithmetic, dependencies, citations, and archive agree with the manuscript. These do not constitute candidate completeness beyond the declared registries.

Exact nonclaim boundary

The paper does not derive the experimentally realized global form; prove that $\Gamma = \mathbb{Z}_6$ is uniquely selected; infer a unique maximal dyonic branch from the gauge group alone; identify the gauge-topological theta torus with the complete physical phase space; classify every spin/nonspin refinement, discrete theta angle, noncompact background, or ultraviolet monopole completion; assign exact higher-flavor symmetry to the generic nonzero-Yukawa Standard Model; extend the massless noninvertible defects to massive QED/QCD without deformation analysis; determine numerical theta angles or couplings; or discharge the whole Standard Model symmetry exhaustion reserved for SYM-11.

One-sentence falsifier

The principal theorem fails if, at a fixed target fingerprint and operator category, the stated common center, quotient family, line descent, selected maximal mutually local branch, screening quotient, one-form group, theta-identification torus, physical phase quotient, background-action/fractionalization data, higher-group interface, noninvertible target boundary, or presentation covariance is wrong; or if two packets claimed equivalent differ on an admitted genuine extended, background-response, or topological observable. A different ultraviolet completion, operator category, monopole policy, line branch, fractionalization class, Yukawa limit, or spin/background structure is not a same-target counterexample unless a transport theorem preserves all of those fields.

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I

Scope, proof authority, and the local-to-global problem

1 Arc position and the strengthened SYM-9 endpoint

1.1 The problem hidden by the notation $SU(3) \times SU(2) \times U(1)$

The expression conventionally used for the Standard Model gauge group often denotes only its local Lie algebra and a convenient product cover. That shorthand is sufficient for most perturbative Feynman rules, but it does not specify the global topology of the gauge group, the representations available to heavy probe lines, the magnetic charge lattice, the possible bundle classes, or the identifications of topological angles. Gauge theories with the same Lie algebra and the same local perturbative particle content can differ in precisely those data. The difference is invisible if the operator category is restricted to point-local perturbative fields, but becomes physical when line operators, monopoles, nontrivial bundles, or background generalized-symmetry fields are admitted [1, 2, 3].

The central question of SYM-9 is therefore not merely “what quotient notation should be written under the Standard Model?” It is:

What global gauge and extended-operator data must be fixed before the Standard Model becomes a determinate global quantum field-theory target, and which changes are lawful presentations of one target rather than changes of target?

The answer requires a simultaneous treatment of the common center action, quotient group, a chosen maximal mutually local line-spectrum branch, dynamical screening, one-form symmetry, theta periodicity, background-action/fractionalization data, higher-group structure, and noninvertible defect fibers.

1.2 Relation to SYM-0 through SYM-8

The preceding papers supply a typed local foundation rather than the SYM-9 conclusion.

SYM-0 fixes the symmetry-status record and the distinction among exactness, physical quotient action, and structural role [4]. SYM-1 separates the load-bearing gauge architecture from compact-support local gauge operations and explicitly leaves global form and line sectors open [5]. SYM-3 supplies the anomaly vocabulary required to distinguish perturbative gauge inconsistency, global anomaly, and generalized-symmetry matching [6]. SYM-5 supplies the zero-Yukawa flavor target and basis/spurion firewall used in the higher-group section [7]. SYM-7 supplies operator-dimension and neutrino-target discipline for global baryon/lepton claims [8]. SYM-8 fixes the one-doublet electroweak residual generator and the point-particle charge packet, while deferring the faithful global electromagnetic subgroup and monopole/line sectors [9].

Accordingly, the present paper does not reopen local gauge quotient, flavor-basis, anomaly, or electroweak-fixation results. It tests what survives when the operator category is enlarged from local fields to extended probes and when the product cover is replaced by an actual global group.

1.3 Strengthening relative to the arc plan

The original arc plan assigns SYM-9 seven burdens: local/global separation, line-operator classification, one-form symmetry, theta periodicity, higher flavor, noninvertible structures, and a closure-scope theorem [10]. The present paper preserves all seven and strengthens their composition.

The strengthening is a *normal-form theorem at a complete fixed global target*. The global form fixes electric

descent but does not, in complete generality, select a unique maximal mutually local magnetic/dyonic line spectrum. Once Γ , the line-spectrum branch $\mathcal{P}_\Gamma$, the operator category, magnetic-sector policy, background transformation package, fractionalization or extension class F , and relevant mass/limit data are fixed, the screening quotient, one-form symmetry, gauge-topological theta torus, physical phase quotient, and admitted generalized-symmetry responses cannot be selected independently. They are components of one common global completion. A proposed package that changes one of these while holding the rest fixed is either inconsistent or describes a different target.

Strengthened endpoint

For each admissible $\Gamma \leq K_{\text{mat}}$ and complete target fingerprint, the completed Standard Model target carries a jointly constrained packet

$$\mathbb{G}_\Gamma = (G_\Gamma, \mathcal{P}_\Gamma, \mathcal{L}_\Gamma^P, \mathcal{S}_\Gamma, \mathcal{M}_\Gamma, \mathcal{A}_\Gamma^{(1)}, \Theta_\Gamma^{\text{line}}, \Phi_\Gamma^{\text{phys}}, \mathcal{F}_\Gamma, \mathcal{H}_\Gamma^{\text{higher}}, \mathcal{N}_\Gamma).$$

The packet is invariant under lawful presentation changes. Two packets that differ on a genuine extended-operator, background-action, gauge-topological, or physical-phase observable are different completed targets. The paper classifies the family of such normal forms; it does not select one global form or line-spectrum branch as uniquely realized by nature.

1.4 Why the family result is not a weakened uniqueness claim

The failure to select a unique Γ or a unique line-spectrum branch from the global group alone is not an incompleteness in the theorem. The local Standard Model data simply do not contain the information required to make either selection. Those are additional global target data. A theorem that invented a unique quotient or line polarization from the local algebra would confuse underdetermination with derivation.

The family theorem is stronger in a different direction. It says that each fully specified global target—including Γ , line branch, monopole policy, and applicable background-action data—has determinate consequences for the admitted extended operator and topological structure. The freedom lies in choosing the global target; it does not persist as arbitrary freedom inside a fixed target.

1.5 Principal results

The analysis establishes three principal results.

1. The local Standard Model does not specify a unique global gauge theory.
2. The four standard global-form targets, once completed by their declared line branch and background-action data, have systematically different line, one-form, monopole, theta-identification, physical-phase, and response packets.
3. Higher-group and noninvertible symmetry statements are target-fiber claims whose exact assumptions must remain visible.

The AASC contribution is to compose these field-theory facts into a redescription-stable role map and a precise downstream closure boundary.

2 Fixed targets, operator categories, and exact nonclaims

2.1 The local Standard Model target

Definition 2.1 (Local target $D_{\text{SM}}^{\text{loc}}$). The local target $D_{\text{SM}}^{\text{loc}}$ is the four-dimensional spin quantum field theory on a fixed oriented Minkowski background with local gauge algebra

$$\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_{\widetilde{Y}},$$

three generations of the minimal chiral Standard Model fermion representations, one Higgs doublet, the renormalizable interaction grammar, and the perturbative gauge-anomaly cancellation ledger. Hypercharge is normalized integrally by $\tilde{Y} = 6Y$. The local operator category contains gauge-invariant point-local operators, ordinary perturbative states, and the local connection/curvature data required for the field theory.

For one generation the left-handed Weyl fields and Higgs doublet are recorded as

$$Q : (\mathbf{3}, \mathbf{2})_{+1},$$

$$u^c : (\bar{\mathbf{3}}, \mathbf{1})_{-4}, \quad d^c : (\bar{\mathbf{3}}, \mathbf{1})_{+2}, \quad L : (\mathbf{1}, \mathbf{2})_{-3}, \quad e^c : (\mathbf{1}, \mathbf{1})_{+6}, \quad H : (\mathbf{1}, \mathbf{2})_{+3}.$$

No right-handed neutrino is silently imported. The same convention is used in [table 1](#) and throughout the global-form calculation.

2.2 Global-form targets

Let

$$G_0 = \mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_{\tilde{Y}}$$

be the reference product cover. A global-form target is not defined until a subgroup Γ of the common representation kernel has been chosen.

Definition 2.2 (Global-form target). For $\Gamma \leq K_{\mathrm{mat}}$, define

$$G_\Gamma = G_0/\Gamma.$$

The target $D_{\mathrm{SM},\Gamma}^{\mathrm{loc}}$ consists of $D_{\mathrm{SM}}^{\mathrm{loc}}$ together with the principal G_Γ -bundle class and the representation descent of the local matter package.

The word “faithful” is used carefully. The quotient $G_{\mathbb{Z}_6}$ acts faithfully on the listed local matter package. For smaller Γ , the remaining subgroup K_{mat}/Γ acts trivially on the local dynamical matter and appears as an electric one-form symmetry once the line category is admitted. Those smaller-quotient theories remain legitimate global gauge theories because heavy Wilson probes can transform under representations unavailable to the local matter fields. They must not be discarded merely because the local matter representation is nonfaithful.

2.3 Operator-category ladder

The paper distinguishes three nested operator scopes.

Definition 2.3 (Operator categories). 1. $\mathcal{O}_{\mathrm{loc}}$ contains point-local gauge-invariant operators and the perturbative particle sector.

2. $\mathcal{O}_{\mathrm{line}}$ adjoins Wilson, ’t Hooft, and dyonic line operators, with global descent and mutual locality imposed.

3. $\mathcal{O}_{\mathrm{def}}$ further adjoins the background fields, topological surfaces, defects, and junction data required for higher-form, higher-group, or noninvertible symmetry statements.

The relation $\mathcal{O}_{\mathrm{loc}} \subset \mathcal{O}_{\mathrm{line}} \subset \mathcal{O}_{\mathrm{def}}$ is a relation among target specifications, not a claim that every physical experiment accesses every operator. A theory may be locally indistinguishable from another at $\mathcal{O}_{\mathrm{loc}}$ and nevertheless differ at $\mathcal{O}_{\mathrm{line}}$.

2.4 Line-polarization and theta-branch data

A global gauge group does not, in complete generality, select a unique maximal mutually local magnetic/dyonic line spectrum. Distinct consistent choices can define distinct theories with the same local correlators, and theta shifts can permute such choices [1]. The Standard Model line analysis must therefore carry the selected branch rather than hide it inside the phrase “theta frame.”

Definition 2.4 (Line-polarization/theta-branch datum). At fixed G_Γ , dynamical matter content, and magnetic policy, $\mathbf{P}$ consists of:

1. a reference class $[\mathfrak{D}_0] \in \Theta_\Gamma^{\text{line}}$ in the gauge-topological line torus;
2. a chosen maximal DSZ-mutually-local dyonic charge lattice $\Lambda_{\text{dy},\Gamma}^{\mathbf{P}}$ containing the local dynamical charge lattice and compatible with electric descent;
3. its Witten-effect transport ρ_Θ under the theta-identification lattice.

Two such data define the same branch when a pairing-preserving lattice isomorphism, together with theta-lattice transport, identifies them. Branches not related in this way are distinct line-completed targets.

The concrete Standard Model tables use the reference branch analyzed in [2], specified at a reference theta class and transported by the displayed theta lattice. The paper does not claim that G_Γ and DSZ compatibility alone select a unique maximal mutually local line branch.

2.5 Magnetic-sector policies

A magnetic one-form symmetry exists only relative to a declared dynamical monopole policy. We use the following values.

Definition 2.5 (Magnetic policy). 1. M_0 : no dynamical hypercharge monopoles are admitted; ‘t Hooft lines are probes and the Bianchi current is conserved.

2. M_Λ : a specified lattice Λ_{mon} of dynamical magnetic charges is admitted, and the magnetic one-form symmetry is reduced to the character group annihilating that lattice.
3. M_{UV} : a named ultraviolet completion supplies monopoles and may alter the global-form and flavor/higher-group analysis. This is a target update unless the ultraviolet data are part of the theorem statement.

2.6 Flavor, background, and massless fibers

The generic observed Standard Model target contains nonzero Yukawa matrices and nonzero charged-fermion masses after electroweak fixation. Higher flavor and noninvertible claims require neighboring fibers:

- $D_{Y=0}^{\text{bg}}$: the zero-Yukawa flavor limit, with background flavor and magnetic one-form gauge fields admitted;
- $D_{\text{QED}}^{m_e=0,\text{def}}$: massless QED with the topological defect category of [11];
- $D_{\text{QCD}}^{m_u=m_d=0,\text{IR}}$: first-generation massless QCD below the electroweak scale, with its pion effective target and matching data.

No conclusion proved only in these fibers is written as an exact symmetry of the generic massive core.

2.7 Background-action and fractionalization data

The list of admitted background fields does not always determine how a lower-form symmetry acts on objects charged under a higher-form symmetry. That action can require an additional fractionalization or Postnikov-type class.

Definition 2.6 (Background-action/fractionalization datum). The coordinate F records the additional data needed to specify the admitted background symmetry action: a fractionalization or Postnikov class, junction decoration, and the allowed response-counterterm class where applicable. For the $B - L$ application with discrete electric one-form factor

$$\mathcal{A}_{\text{el},\Gamma}^{(1)} \cong \mathbb{Z}_6/\Gamma,$$

the action on charged lines can require

$$k \in H^2(BU(1)_{B-L}, \mathbb{Z}_6/\Gamma),$$

and the resulting global-form-sensitive transport depends on that choice [12]. If no mixed zero-/one-form background structure is admitted, F is marked not applicable.

Thus B records which backgrounds are present, whereas F records how the admitted symmetry acts in their presence. Two targets may have the same background-field set and different F data.

2.8 Target fingerprint

Target fingerprint

Every theorem in the paper is indexed, explicitly or by inherited section convention, by

$$(D, \Gamma, O, M, P, B, F, Y, S),$$

where D is the field theory, Γ the global form, O the operator category, M the monopole policy, P the line-polarization/theta branch, B the background-field set, F the background-action/fractionalization class, Y the Yukawa/mass limit and retained phase package, and S the spin/nonspin geometric scope. Altering one of these fields is a target update unless a theorem transports the relevant structures.

2.9 Exact nonclaims

The paper does not claim that the four-target family exhausts disconnected gauge groups, noncompact forms, or all spin-charge refinements. It does not decide which global form is realized experimentally. It does not treat a heavy probe line as a dynamical particle, or the existence of a probe as an experimental observation. It does not infer a generalized symmetry from a center subgroup without screening and monopole analysis. It does not identify the hypercharge theta coordinate with an observable in the absence of magnetic or boundary probes. It does not claim that the gauge group and DSZ condition alone select a unique maximal line polarization. It does not infer a background symmetry action from the background-field list when a fractionalization class is required. It does not identify the gauge-topological torus $\Theta_{\Gamma}^{\text{line}}$ with the complete physical phase packet $\Phi_{\Gamma}^{\text{phys}}$, or infer full Standard Model CP or time-reversal invariance from a fixed class of $\Theta_{\Gamma}^{\text{line}}$. It does not claim that every massless-limit noninvertible defect survives the physical fermion masses. It does not replace SYM-11's whole-atlas candidate completeness theorem.

3 Imported interfaces and proof authority

3.1 Authority classes

The manuscript uses four authority classes and keeps them distinct.

1. **Locally proved:** center action, subgroup classification, electric descent congruences, the selected mutual-locality calculations, screening consequences, and the integrated normal-form propositions proved in this paper.

2. **Primary-literature imports:** the classification of line operators for the Standard Model global forms, the possible nonuniqueness of line spectra at fixed gauge group, Standard Model dyonic-lattice and physical-theta classification, generalized-symmetry formalism, symmetry fractionalization, 2-group background laws, higher-flavor application, and massless noninvertible defects [2, 1, 13, 3, 12, 14, 15, 11].
3. **Upstream AASC interfaces:** target, quotient, transport, representation, charge, anomaly, holonomy, and phase-role results already established in the corpus.
4. **Conditional target inputs:** monopole spectrum, line-spectrum branch or discrete theta-like choice, background fields and fractionalization/extension class, Yukawa limit, and defect completion when the paper studies a neighboring target rather than the generic core.

A statement does not become locally proved because it is repeated in the data files or because its citation is stable.

3.2 Gauge and transport imports

The Gauge Transition Groupoid, connection, curvature, Lie-type realization, and product-compatibility manuscripts establish the boundary-fixed local transition and transport architecture [16, 17, 18, 19, 20]. Their role in SYM-9 is negative as well as positive: they show that local connection and curvature data are coherent and load-bearing, but they do not determine the faithful global group or the extended-operator spectrum.

Gauge-realization interface
SYM-9 imports a well-defined local product gauge architecture and the distinction between local presentation changes and physical quotient content. It does not import a global quotient, line lattice, monopole spectrum, or theta periodicity from those sources. Those are the new SYM-9 burdens.

3.3 Representation and charge imports

Representation Admissibility, Representation Algebra Specialization, Abelian Charge-Witness Normalization, and the Standard-Model Charge Lattice provide the typed matter representations and the normalization firewall [21, 22, 23, 24]. In particular, a rescaling of the Abelian generator and inverse rescaling of its coupling is a lawful coordinate change; charge ratios and the common central kernel are transported rather than recreated.

These imports authorize the center calculation to be performed on the frozen representation table. They do not authorize the inference that one quotient is physically selected simply because it is the maximal quotient invisible to local matter.

3.4 Anomaly imports

The anomaly manuscripts distinguish perturbative local gauge anomaly cancellation from global congruence, index, and matching questions [25, 26, 6]. SYM-9 imports the anomaly-neutral local Standard Model architecture. It separately records that global-form choices can alter global anomaly questions and that a complete global target must include the relevant spin/bundle assumptions [27].

No claim in the paper uses perturbative anomaly cancellation as a proof that the global gauge group is unique.

3.5 Holonomy and phase imports

Discrete Holonomy Exactness, CP-like Phases, and the Strong-CP Boundary Endpoint provide a representative/invariant discipline for finite holonomy and theta coordinates [28, 29, 30]. They are used to block two mistakes:

1. treating a coordinate shift of a theta angle as a new physical sector without checking the global identification lattice;
2. treating a Wilson-loop or holonomy display as a genuine line-operator classification without the global representation and screening analysis.

The actual Standard Model theta-identification lattice is taken from the line-operator analysis and rederived in the chosen convention.

3.6 Primary field-theory sources

Tong supplies the concrete four-form global family, a definite maximal mutually local line branch for each quotient at a reference theta class, electric/magnetic line congruences, theta ranges and coupled shifts, and electroweak-broken charge examples [2]. Aharony, Seiberg, and Tachikawa supply the stronger general lesson that fixing a gauge group need not uniquely fix the maximal mutually local magnetic/dyonic line spectrum: different branches can be permuted by ordinary theta shifts or labeled by discrete theta-like data [1]. Alonso, Chadha-Day, Dimakou, Ha, and Khoze provide a 2026 Standard Model dyonic-lattice classification, quotient-dependent theta domains and CP points, and an explicit reduction of the three gauge-theta coordinates to two physical directions after the anomalous $B + L$ quotient [13]. Gaiotto, Kapustin, Seiberg, and Willett supply the higher-form symmetry language [3]. Córdova, Dumitrescu, and Intriligator and Benini, Córdova, and Hsin supply the 2-group background and anomaly formalism [14, 31]. Córdova and Koren supply its zero-Yukawa Standard Model flavor realization and the source-coordinate relation $\widehat{\kappa}_F^{\text{CK}} = -\kappa_F^{\text{CK}}/2$; SYM-9 separately transports the current, two-form background, period lattice, and coefficient to the quotient-unit magnetic-current convention [15]. Choi, Lam, and Shao supply the massless-QED/QCD noninvertible defects [11]. Hsin and Gomis provide an explicit global-form-sensitive topological response involving $B - L$ and the magnetic one-form symmetry, and show that the lower-form action also requires a fractionalization class $k \in H^2(BU(1), \mathbb{Z}_6/\Gamma)$ [12].

3.7 No-circularity rule

The physical theorem must remain readable if the reader rejects the AASC role vocabulary. Conversely, the AASC role map may rely on the established physical facts. The permitted order is

$$\text{target and QFT theorem} \longrightarrow \text{invariant/quotient certificate} \longrightarrow \text{structural role.}$$

A role label never appears in the premise of the center, line, one-form, theta, fractionalization, higher-group, or noninvertible derivation.

4 Notation, normalization, and convention locks

4.1 Integer hypercharge

Throughout the global-form calculation we use

$$\widetilde{Y} = 6Y,$$

so every local matter field carries an integer $q \in \mathbb{Z}$. A hypercharge group element is written $e^{i\alpha}$, and a field of charge q transforms by $e^{iq\alpha}$. This convention makes the common center and line-lattice congruences integral.

A simultaneous rescaling $\widetilde{Y} \mapsto r\widetilde{Y}$ and $g_Y \mapsto g_Y/r$ is a coordinate change only if the character lattice and the period of the $U(1)$ parameter are transported with it. The fixed integer convention avoids such rescalings inside the main proof.

4.2 Center labels

Let

$$\omega_3 = e^{2\pi i/3}.$$

For an $SU(3)$ representation, $z_3 \in \mathbb{Z}_3$ denotes its triality; for an $SU(2)$ representation, $z_2 \in \mathbb{Z}_2$ denotes its center parity. The fundamental representations have $z_3 = 1$ and $z_2 = 1$, while the anti-fundamental of $SU(3)$ has $z_3 = -1$ modulo 3.

The central generator used in the local calculation is

$$\xi = (\omega_3, -1, e^{i\pi/3}).$$

Replacing ξ by another generator of the same embedded cyclic subgroup does not change the quotient target.

4.3 Electric labels

An electric probe line is labeled by

$$\ell_e = (z_{2,e}, z_{3,e}, q), \quad z_{2,e} \in \mathbb{Z}_2, \quad z_{3,e} \in \mathbb{Z}_3, \quad q \in \mathbb{Z}.$$

These are center charges rather than complete highest weights. They suffice for the quotient and mutual-locality congruences. Different representations with the same center charges may differ in local Casimir data, but they occupy the same class for the global-form questions addressed here.

4.4 Magnetic and dyonic labels

A magnetic label is

$$\ell_m = (z_{2,m}, z_{3,m}, g), \quad g \in \frac{1}{6}\mathbb{Z}$$

with the allowed fractions determined by Γ . A dyonic line is the pair $\ell = (\ell_e; \ell_m)$. For two lines ℓ and ℓ' , the antisymmetric Dirac–Schwinger–Zwanziger pairing in the chosen normalization is

$$\begin{aligned} \langle \ell, \ell' \rangle &= 3(z_{2,e}z'_{2,m} - z'_{2,e}z_{2,m}) + 2(z_{3,e}z'_{3,m} - z'_{3,e}z_{3,m}) \\ &\quad - 6(qg' - q'g) \in 6\mathbb{Z}. \end{aligned} \tag{1}$$

This is the discrete-center form of mutual locality used in [2].

4.5 Screening and genuine line terminology

The paper keeps three notions separate.

1. A line is *globally admissible* if its electric representation descends to G_Γ and its magnetic data obey the bundle/mutual-locality condition.
2. A line is *genuine* if it is defined without an auxiliary surface in the chosen global theory.
3. Two genuine lines are *screening equivalent* if their charges differ by the charge of admitted dynamical matter or dynamical monopoles.

The one-form symmetry acts on the resulting unscreened charge classes. A globally admissible line may still be screenable.

4.6 Theta coordinates and physical phases

We use

$$(\theta_2, \theta_3, \tilde{\theta})$$

for the $SU(2)$, $SU(3)$, and integer-normalized hypercharge theta coordinates. At a fixed line branch and fixed matter-phase convention, their gauge-topological identification torus is

$$\Theta_\Gamma^{\text{line}} = \mathbb{R}^3 / \Lambda_{\theta, \Gamma, \mathcal{P}_\Gamma}.$$

This is not the complete physical phase space of the renormalizable Standard Model. If $\mathcal{Y}_{\text{ph}}$ denotes the retained mass/Yukawa phase data and $\mathcal{R}_{\text{anom}}$ the admitted anomalous fermion rephasings, the physical phase packet is

$$\Phi_{\Gamma}^{\text{phys}} = (\Theta_{\Gamma}^{\text{line}} \times \mathcal{Y}_{\text{ph}}) / \mathcal{R}_{\text{anom}}.$$

The distinction prevents a gauge-topological line-periodicity statement from being promoted into a claim about all physically inequivalent CP phases.

4.7 One-form notation

The electric one-form group is written $\mathcal{A}_{\text{el},\Gamma}^{(1)}$ and the magnetic group $\mathcal{A}_{\text{mag},\Gamma}^{(1)}$. In the no-dynamical-monopole target,

$$\mathcal{A}_{\text{mag},\Gamma}^{(1)} \cong \text{U}(1),$$

with current normalized as

$$J_{\Gamma}^{(2)} = \star \frac{n \, da_Y}{2\pi}, \quad n = |\Gamma|.$$

This normalization makes the minimally allowed magnetic probe carry unit one-form charge [12].

4.8 Higher-group coefficient normalization

The numerical 2-group coefficient is always quoted together with a magnetic-current and two-form-background normalization. The Córdoba–Koren source coordinate satisfies

$$\widehat{\kappa}_F^{\text{CK}} = -\frac{\kappa_F^{\text{CK}}}{2}.$$

The one-form theorem uses the quotient-unit current $J_{\Gamma}^{(2)} = n \star da_Y / (2\pi)$. If $J_{\Gamma}^{(2)} = r_{\Gamma} J_{\text{CK}}^{(2)}$, the contragredient background coordinate and coefficient satisfy

$$B_{\Gamma}^{(2)} = r_{\Gamma}^{-1} B_{\text{CK}}^{(2)}, \quad \widehat{\kappa}_{F,\Gamma} = r_{\Gamma}^{-1} \widehat{\kappa}_F^{\text{CK}},$$

with the two-form period lattice rescaled simultaneously. When $J_{\text{CK}}^{(2)}$ is identified with the unquotiented current $\star da_Y / (2\pi)$, one has $r_{\Gamma} = n$. The invariant datum is

$$\mathcal{K}_{F,\Gamma} = [J_{\Gamma}^{(2)}, B_{\Gamma}^{(2)}, \text{Per}(B_{\Gamma}^{(2)}), \widehat{\kappa}_{F,\Gamma}]_{\text{norm}},$$

not a bare coefficient copied between conventions.

4.9 Baryon-minus-lepton background normalization

Whenever the Hsin–Gomis response is used, we retain its zero-form charge convention explicitly. Let B_{q} denote the quark-number generator, which assigns +1 to Q_i and −1 to $\bar{u}_i, \bar{d}_i$. Then

$$Q_{B-L}^{\text{HG}} = \frac{1}{3} B_{\text{q}} - \sum_{i=e,\mu,\tau} L_i.$$

Equivalently, if $B_{\text{conv}} = B_{\text{q}}/3$ is conventional baryon number, then $Q_{B-L}^{\text{HG}} = B_{\text{conv}} - \sum_i L_i$. Quark fields therefore carry projective charges in thirds, while minimally charged gauge-invariant local operators have unit charge. The numerical fractional response $\xi_{\text{HG}}(\Gamma, k)$ is not transported independently of this zero-form charge convention.

4.10 Stable theorem identifiers

The principal statements carry the stable identifiers SYM9-F01–SYM9-Q13, SYM9-A01–SYM9-A04, and SYM9-MAIN. The identifiers are independent of displayed theorem numbering and are frozen in the theorem registry.

II

Global-form family and extended operator reconstruction

5 Local-to-global underdetermination

5.1 The same Lie algebra supports different global groups

A Lie algebra determines infinitesimal transformations and local curvature, but not the global topology of the gauge group. If $\tilde{G}$ is the simply connected semisimple cover together with the chosen Abelian factor, any subgroup of the center acting trivially on the admitted local representations may be quotiented without changing the local algebra. The resulting theories have the same perturbative gauge bosons and local covariant derivatives, yet their principal bundles and extended operators differ.

For the Standard Model reference cover G_0 , the quotient choice is invisible to the local Lagrangian because every dynamical field representation factors through all four groups G_Γ derived below. The covariant derivative

$$D_\mu = \partial_\mu + A_\mu^a T_a + B_\mu \tilde{Y}$$

contains only the Lie algebra action. It does not record which center elements are identified with the identity in the global group.

Theorem 5.1 (Local-to-global underdetermination, SYM9-Q01). *Fix $D_{\text{SM}}^{\text{loc}}$ and restrict observables to $\mathcal{O}_{\text{loc}}$. The local gauge algebra, perturbative connection and curvature, local matter representation differentials, and perturbative anomaly-cancellation equations do not determine a unique compact connected global gauge group. If Γ_1 and Γ_2 are distinct subgroups of the common central kernel, then G_{Γ_1} and G_{Γ_2} induce the same local Lie-algebra action on the frozen matter fields but may differ in principal-bundle topology, globally admissible Wilson lines, magnetic sectors, one-form symmetry, and theta identifications.*

Proof. The quotient map $G_0 \rightarrow G_\Gamma$ is a local diffeomorphism, so each G_Γ has Lie algebra $\mathfrak{g}_{\text{SM}}$. Every frozen local matter representation is trivial on Γ by definition of the common kernel; therefore it descends to a representation of G_Γ with the same differential. Local gauge-covariant operators and perturbative vertices depend on that differential and are unchanged.

The converse fails because a representation of the Lie algebra need not integrate to every representation of a chosen global group. A heavy Wilson probe must be a representation of G_Γ , hence must be trivial on Γ . Magnetic charges are cocharacters of the global group, and their allowed lattice changes contragrediently with the electric character lattice. Nontrivial bundles and topological charges likewise depend on $\pi_1(G_\Gamma)$ and on the quotient embedding. Thus the listed local data are common to the family while the extended data need not be. The explicit differences are derived in sections 8, 10 and 13. $\square$

5.2 Local equivalence is operator-category relative

The theorem is not the assertion that the four theories are physically identical. It says that the local operator category does not separate them. Once $\mathcal{O}_{\text{line}}$ is admitted, the equivalence relation becomes finer.

Let $\mathcal{A}_{\text{loc}}(G_\Gamma)$ denote the local observable algebra and $\mathcal{A}_{\text{ext}}(G_\Gamma)$ the algebra or category enlarged by genuine lines and their junctions. It is possible that

$$\mathcal{A}_{\text{loc}}(G_{\Gamma_1}) \cong \mathcal{A}_{\text{loc}}(G_{\Gamma_2})$$

while

$$\mathcal{A}_{\text{ext}}(G_{\Gamma_1}) \not\cong \mathcal{A}_{\text{ext}}(G_{\Gamma_2}).$$

This is the sense in which distinct global forms can be “the same local theory” but different completed theories [1].

5.3 Why anomaly cancellation does not select the quotient

Perturbative gauge anomalies are local functionals of the Lie-algebra representations. They must cancel for every admissible global completion, but their cancellation does not determine the global quotient. Global anomalies can depend on the global form and spacetime structure, yet they are additional tests, not a mechanism that generically collapses the four-form family to one member [27].

Adversarial check

A common invalid inference is

$$\text{local anomaly-free matter representation} \implies \text{unique global gauge group}.$$

The premise fixes the infinitesimal consistency ledger. It does not fix the character and cocharacter lattices or the allowed bundle topology. Any uniqueness claim must identify a separate global datum or observation.

5.4 Operational consequence

SYM-9 must therefore avoid two opposite errors. It may not treat Γ as pure notation, because line and topological data change. It may not treat the largest quotient as uniquely forced, because smaller quotients are consistent global targets with additional heavy-probe sectors and electric one-form symmetry. The correct object is a family indexed by Γ and operator scope.

6 The common center action of the Standard Model matter package

6.1 Central action formula

Let a field carry $SU(3)$ triality $t \in \mathbb{Z}_3$, $SU(2)$ center parity $s \in \mathbb{Z}_2$, and integer hypercharge q . The element

$$\xi = (\omega_3, -1, e^{i\pi/3})$$

acts by

$$\rho_{t,s,q}(\xi) = \omega_3^t (-1)^s e^{i\pi q/3} = \exp \left[\frac{i\pi}{3} (2t + 3s + q) \right].$$

Therefore ξ is invisible on a representation exactly when

$$2t + 3s + q \equiv 0 \pmod{6}.$$

6.2 Kernel theorem

Theorem 6.1 (Common-center kernel, SYM9-Q02). *For the frozen Standard Model matter and one-Higgs-doublet representation package in integer hypercharge normalization, the kernel of the product-cover action is*

$$K_{\text{mat}} = \langle \xi \rangle \cong \mathbb{Z}_6, \quad \xi = (\omega_3, -1, e^{i\pi/3}).$$

Equivalently, the local matter action factors through $G_0/\mathbb{Z}_6$, and every subgroup of K_{mat} is also invisible to the local matter fields.

Table 1: Action of the common center generator on one Standard Model generation and the Higgs. Triality is written as 1, -1, 0 and weak parity as 1, 0.

Field	$SU(3)$	$SU(2)$	t	$q = \tilde{Y}$	$2t + 3s + q \pmod{6}$
Q	3	2	1	1	0
u^c	$\bar{\mathbf{3}}$	1	-1	-4	0
d^c	$\bar{\mathbf{3}}$	1	-1	2	0
L	1	2	0	-3	0
e^c	1	1	0	6	0
H	1	2	0	3	0

Proof. The entries in table 1 verify that ξ acts trivially on every irreducible multiplet. Its order is six because the three components have orders 3, 2, and 6, and no positive power smaller than six is simultaneously the identity in all three factors.

It remains to show that the kernel contains no larger connected or finite subgroup. Let $z = (\omega_3^a, (-1)^b, e^{i\alpha})$ act trivially. Triviality on e^c gives $e^{6i\alpha} = 1$, so $\alpha = k\pi/3$ modulo 2π . Triviality on L gives $(-1)^b e^{-3i\alpha} = 1$, hence $b \equiv k \pmod{2}$. Triviality on d^c gives $\omega_3^{-a} e^{2i\alpha} = 1$, hence $a \equiv k \pmod{3}$. Therefore

$$z = (\omega_3^k, (-1)^k, e^{ik\pi/3}) = \xi^k.$$

Thus the full kernel is exactly $\langle \xi \rangle$. □

6.3 Normalization covariance

The calculation is independent of the displayed hypercharge unit. If $\tilde{Y}$ is replaced by $r\tilde{Y}$ while the $U(1)$ period and coupling are transported consistently, the same embedded subgroup is described by a different Abelian phase coordinate. The invariant object is the kernel of the representation homomorphism, not the integer tuple used to display it.

Proposition 6.2 (Normalization covariance, SYM9-F02). *A lawful hypercharge normalization change transports the character lattice, Abelian period, coupling, center generator, line charges, and theta coordinate together. It preserves the isomorphism class and embedded action of K_{mat} , the quotient family, the electric descent relations, and all charge ratios.*

Proof. A normalization change is an automorphism of the one-dimensional Lie algebra accompanied by an inverse coupling rescaling and a relabeling of the character lattice. The representation homomorphism as a map into phases is unchanged. Hence its kernel and every quotient by a subgroup of that kernel are unchanged up to canonical isomorphism. The line and theta coordinates are dual lattice coordinates and transform contragrediently. □

6.4 The maximal local-matter quotient and the family of global targets

The theorem explains two statements that are often conflated.

1. $G_{\mathbb{Z}_6}$ is the quotient of the reference cover that acts faithfully on the listed local dynamical matter.
2. G_1 , $G_{\mathbb{Z}_2}$, and $G_{\mathbb{Z}_3}$ are distinct global gauge theories compatible with the same local matter, because the residual kernel acts as a one-form symmetry on heavy Wilson probes rather than as a new local particle action.

The first does not eliminate the second.

7 The compact connected global-form family

7.1 Subgroup classification

A cyclic group of order six has one subgroup of each order dividing six. Hence

$$\Gamma \in \{1, \langle \xi^3 \rangle, \langle \xi^2 \rangle, \langle \xi \rangle\} \cong \{1, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_6\}.$$

Explicitly,

$$\xi^3 = (1, -1, -1), \quad \xi^2 = (\omega_3^2, 1, e^{2\pi i/3}).$$

The choice of inverse generator for the $\mathbb{Z}_3$ subgroup is a presentation change.

Theorem 7.1 (Compact connected global-form family, SYM9-Q03). *Within the target class of compact connected gauge groups obtained from G_0 by quotienting a central subgroup invisible to the frozen local Standard Model matter representation, the complete family is*

$$G_1, \quad G_{\mathbb{Z}_2}, \quad G_{\mathbb{Z}_3}, \quad G_{\mathbb{Z}_6}.$$

No fifth quotient occurs in this declared family.

Proof. By theorem 6.1, every allowed quotient subgroup lies in $K_{\text{mat}} \cong \mathbb{Z}_6$. The subgroup lattice of a cyclic group has precisely one subgroup for each divisor of its order. The divisors of six are 1, 2, 3, 6. Conversely, quotienting by any of these subgroups is compatible with every frozen local matter representation because the subgroup lies in the kernel. Thus the four targets are necessary and sufficient inside the declared quotient family. $\square$

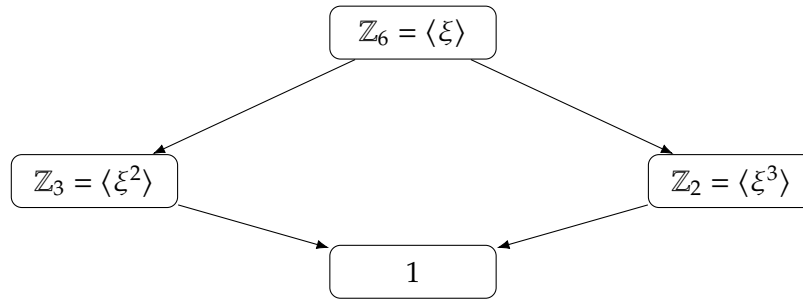


Figure 1: Subgroup lattice of the common Standard Model matter kernel. The arrows point toward smaller quotient subgroups, not toward a dynamically preferred global form.

7.2 Target identity and quotient order

Write $n = |\Gamma|$. The quotient order controls several later formulas, but n alone does not replace the embedded subgroup. In the present cyclic kernel the order uniquely identifies the subgroup; in more general gauge products it need not.

The four targets have the same connected Lie algebra and the same local matter content. Their differences are visible in:

- which heavy electric representations descend;
- which fractional hypercharge magnetic fluxes are admitted;
- the electric one-form group $\mathbb{Z}_{6/n}$;
- the normalization of the magnetic one-form current;
- the coupled theta-identification lattice;
- the electroweak-broken minimum probe charges and color-magnetic content.

7.3 No unique selection theorem

Exact nonclaim boundary

Theorem 7.1 classifies the quotient family. It does not prove that nature realizes $G_{\mathbb{Z}_6}$, that ultraviolet completeness requires the maximal quotient, or that the existence of local fractional charges selects one global group. Such a selection requires an independent observation or ultraviolet theorem sensitive to the extended sector.

7.4 A physical discriminator beyond local correlators

Hsin and Gomis exhibit a global-form-sensitive fractional topological response involving $B - L$ and the magnetic one-form current [12]. Its fractional part can distinguish global forms for suitable fractionalization classes even though the local perturbative gauge vertices agree. This example is not needed to prove the four-form family, but it demonstrates that the family is physically meaningful rather than a notational ambiguity.

8 Electric line operators and global descent

8.1 Wilson lines know the global group

A Wilson line in a representation R of the product cover is globally defined in the quotient theory G_Γ only if every element of Γ acts trivially on R . At the center-charge level, write the electric label as

$$\ell_e = (z_{2,e}, z_{3,e}, q).$$

The common generator acts by

$$\rho_{\ell_e}(\xi) = \exp \left[\frac{i\pi}{3} (3z_{2,e} + 2z_{3,e} + q) \right].$$

If $\Gamma = \mathbb{Z}_n$ is generated by $\xi^{6/n}$, descent requires

$$3z_{2,e} + 2z_{3,e} + q \equiv 0 \pmod{n}.$$

Theorem 8.1 (Electric-line admissibility, SYM9-Q04). *The electric center-charge labels descending to G_Γ obey:*

$$\begin{aligned} \Gamma = 1 : & \text{ no congruence,} \\ \Gamma = \mathbb{Z}_2 : & q \equiv z_{2,e} \pmod{2}, \\ \Gamma = \mathbb{Z}_3 : & q \equiv z_{3,e} \pmod{3}, \\ \Gamma = \mathbb{Z}_6 : & q \equiv 3z_{2,e} - 2z_{3,e} \pmod{6}. \end{aligned}$$

These are necessary and sufficient for the center representation to descend through the corresponding quotient.

Proof. Apply section 8.1. For $n = 2$, the $SU(3)$ term vanishes modulo two and $3z_{2,e} \equiv z_{2,e}$, giving $q + z_{2,e} \equiv 0$, equivalent to the displayed relation. For $n = 3$, the weak term vanishes and $2z_{3,e} \equiv -z_{3,e}$, giving $q \equiv z_{3,e}$. For $n = 6$, rearrangement gives $q \equiv -3z_{2,e} - 2z_{3,e} \equiv 3z_{2,e} - 2z_{3,e}$. Sufficiency follows because the relevant subgroup is generated by $\xi^{6/n}$. $\square$

8.2 The local matter fields occupy the $\mathbb{Z}_6$ descent locus

For every field in table 1, the electric relation

$$q \equiv 3z_2 - 2z_3 \pmod{6}$$

holds. This is the global version of the common-kernel calculation. The relation is not a prediction of the quotient; it is the consistency condition explaining why the local matter representations descend to the maximal quotient.

8.3 Heavy probes distinguish the targets

The smaller-quotient theories admit Wilson lines that are not representations of the larger quotients. For example, a pure hypercharge line of unit q is allowed for $\Gamma = 1$ but fails the $\mathbb{Z}_2$, $\mathbb{Z}_3$, and $\mathbb{Z}_6$ descent conditions. Conversely, the local dynamical matter occupies only the common subset. Heavy probe lines therefore expose global information absent from the local particle spectrum.

8.4 Descent is not screening

A globally admissible Wilson line may be screened by dynamical matter. The descent theorem answers whether the line exists as a representation of the global gauge group. It does not answer whether its long-distance charge labels a one-form symmetry sector. That requires quotienting by the subgroup generated by dynamical matter charges, which is performed in [section 9](#).

Physics translation

The phrase “allowed Wilson line” is ambiguous unless two questions are separated:

1. Is the representation globally defined for G_Γ ?
2. Is its center charge unscreened by the dynamical matter content?

SYM-9 records both fields. The first belongs to $\mathcal{L}_\Gamma^P$; the second to $\mathcal{S}_\Gamma$ and the one-form quotient.

9 Dynamical screening and genuine electric charge classes

9.1 Screening equivalence

Let $\Lambda_{e,\Gamma}^{\text{adm}}$ be the lattice of globally admissible electric line labels and let Λ_e^{dyn} be the subgroup generated by the center charges of admitted dynamical matter. Two electric probes are screening equivalent when their difference lies in Λ_e^{dyn} :

$$\ell_e \sim_{\text{scr}} \ell'_e \iff \ell_e - \ell'_e \in \Lambda_e^{\text{dyn}}.$$

The quotient

$$\overline{\Lambda}_{e,\Gamma} = \Lambda_{e,\Gamma}^{\text{adm}} / \Lambda_e^{\text{dyn}}$$

is the group of unscreened electric line-charge classes. Its Pontryagin dual is the electric one-form symmetry.

9.2 Residual common-center action

Because K_{mat} acts trivially on all dynamical matter, the quotient subgroup not divided out by Γ survives as a symmetry acting on Wilson lines. This gives a direct exact sequence

$$1 \longrightarrow \Gamma \longrightarrow K_{\text{mat}} \longrightarrow K_{\text{mat}}/\Gamma \longrightarrow 1.$$

The remaining group K_{mat}/Γ is the electric one-form symmetry; the line-charge group is its character group.

Theorem 9.1 (Screening quotient, SYM9-Q05). *For the renormalizable Standard Model matter target and global form G_Γ ,*

$$\mathcal{A}_{\text{el},\Gamma}^{(1)} \cong K_{\text{mat}}/\Gamma \cong \mathbb{Z}_{6/n}, \quad n = |\Gamma|.$$

Equivalently, the unscreened electric line-charge group is

$$\overline{\Lambda}_{e,\Gamma} \cong \text{Hom}(\mathbb{Z}_{6/n}, \text{U}(1)) \cong \mathbb{Z}_{6/n}.$$

Thus

$$\begin{array}{c|cccc} \Gamma & 1 & \mathbb{Z}_2 & \mathbb{Z}_3 & \mathbb{Z}_6 \\ \hline \mathcal{A}_{\text{el},\Gamma}^{(1)} & \mathbb{Z}_6 & \mathbb{Z}_3 & \mathbb{Z}_2 & 1. \end{array}$$

Proof. The center subgroup K_{mat} is invisible to all dynamical matter by [theorem 6.1](#). In the quotient theory, elements of Γ are gauge identity, so the residual action on Wilson lines is K_{mat}/Γ . A Wilson line transforms by a character of this finite group. Dynamical matter cannot screen that character because it is neutral under K_{mat} ; conversely, representations differing only by local matter charges define the same character. Hence the unscreened charge group is the Pontryagin dual of K_{mat}/Γ , which is cyclic of order $6/n$ and isomorphic to the group itself. $\square$

9.3 Operator-dimension dependence

The theorem is indexed by the admitted dynamical matter and operator grammar. Adding new matter in a representation nontrivial under K_{mat}/Γ screens part or all of the electric one-form symmetry. A heavy probe does not break the symmetry; a dynamical field that can end the line does. Likewise, a higher-dimension local operator built only from the existing fields does not change the center charges of the dynamical generators, though a target extension with new particles can.

9.4 Screening versus confinement

Screening equivalence is not identical to an area/perimeter law classification. A line can be unscreened as a charge class yet confined in a phase, or screenable through pair creation without being absent as an ultraviolet line operator. The one-form group records the exact topological action on line charges at the specified theory target, while phase behavior is a separate dynamical field.

Adversarial check

The inference

$$\text{center of the gauge group} \implies \text{one-form symmetry}$$

is incomplete. The correct chain is

$$\text{center action} + \text{quotient} + \text{dynamical matter screening} \implies \text{electric one-form symmetry}.$$

The screening step is load-bearing.

10 Magnetic and dyonic line sectors

10.1 Cocharacter data and mutual locality

Electric representations are characters of a maximal torus subject to quotient descent. Magnetic charges are cocharacters, and the two lattices are constrained by the Dirac–Schwinger–Zwanziger pairing. In the center-charge coordinates of [equation \(1\)](#), mutual locality requires

$$\langle \ell, \ell' \rangle \in 6\mathbb{Z}$$

for every pair of genuine lines in the chosen maximal mutually local set.

For a pure electric line and pure magnetic line this becomes

$$3z_{2,e}z_{2,m} + 2z_{3,e}z_{3,m} - 6qg \in 6\mathbb{Z}.$$

As the electric lattice becomes more restricted by a larger quotient, the magnetic lattice becomes less restricted. Mutual locality constrains admissible line spectra, but does not by itself choose one maximal mutually local branch when several are available; that branch is the target coordinate $\mathcal{P}_\Gamma$ of [theorem 2.4](#).

Theorem 10.1 (Magnetic and dyonic compatibility, SYM9-Q06). *At the reference theta class of the line-spectrum branch $\mathcal{P}_{\Gamma, \Gamma}$ fixed in theorem 2.4, the minimal magnetic congruences compatible with the electric descent lattices are:*

$$\begin{aligned}\Gamma = 1 : \quad & z_{2,m} = 0, \quad z_{3,m} = 0, \quad g \in \mathbb{Z}; \\ \Gamma = \mathbb{Z}_2 : \quad & z_{3,m} = 0, \quad g \equiv \frac{z_{2,m}}{2} \pmod{1}; \\ \Gamma = \mathbb{Z}_3 : \quad & z_{2,m} = 0, \quad g \equiv \frac{z_{3,m}}{3} \pmod{1}; \\ \Gamma = \mathbb{Z}_6 : \quad & 6g \equiv 3z_{2,m} + 2z_{3,m} \pmod{6}.\end{aligned}$$

A convenient magnetic basis for the maximal quotient on this branch is

$$(z_{2,m}, z_{3,m}; g) = (1, 0; 1/2), \quad (0, 1; 1/3).$$

Together with the electric lattice and equation (1), these data generate the selected maximal mutually local dyonic charge lattice at the reference theta class.

Proof. The result follows by demanding section 10.1 for every electric label satisfying theorem 8.1. For $\Gamma = 1$, independent unit center charges and unit hypercharge force both non-Abelian magnetic residues to vanish and g to be integral. For $\mathbb{Z}_2$, the electric relation $q \equiv z_{2,e} \pmod{2}$ permits a half-integral Abelian magnetic charge precisely when accompanied by nontrivial $SU(2)$ magnetic residue; independent $SU(3)$ electric charges force $z_{3,m} = 0$. The $\mathbb{Z}_3$ case is analogous. For $\mathbb{Z}_6$, substituting $q = 3z_{2,e} - 2z_{3,e} + 6k$ into section 10.1 and requiring the result for all center residues gives the displayed congruence. The antisymmetric extension equation (1) then supplies the dyonic mutual-locality condition. This reproduces the Standard Model charge-level branch used in [2]. The theorem does not assert that DSZ compatibility and G_Γ alone eliminate every alternative maximal mutually local branch or discrete theta-like refinement [1]. $\square$

10.2 Theta shifts and the Witten effect

At nonzero theta coordinates, a magnetic line generally acquires electric charge. A shift by a theta-lattice generator maps the chosen maximal mutually local line lattice to the corresponding member of its Witten-effect orbit [32]. Therefore the phrase “the magnetic line lattice” is branch- and theta-frame dependent. The invariant object is the selected branch $\mathcal{P}_\Gamma$ together with the theta identification and its monodromy action.

10.3 Dynamical monopoles

The theorem classifies probe lines in the global gauge theory. It does not assert that every allowed magnetic charge is carried by a dynamical particle. If a monopole of charge m is dynamical, lines differing by m can be screened and the magnetic one-form symmetry is reduced. In a gravitational or unified completion, additional completeness assumptions may turn some probe charges into dynamical charges, but that is a separate target premise rather than a conclusion of the low-energy Standard Model.

10.4 Dirac strings are presentation data

A local construction of an ‘t Hooft line uses a singular gauge patch or Dirac string. Moving that string without changing the magnetic charge and mutual-locality class is a presentation change. The magnetic charge class and its action on electric lines are the invariant data.

Locally checkable arithmetic

The accompanying audit recomputes the center-phase formula, the electric descent congruences, and representative DSZ pairings for the displayed magnetic basis. These finite checks protect the published arithmetic for the declared $\mathcal{P}_{\Gamma,T}$ branch; the classification of genuine line operators remains grounded in the field-theory argument and primary line-operator literature.

11 Genuine-line reconstruction and line-sector normal form

11.1 The line packet

The preceding sections supply five logically distinct ingredients:

1. the electric representation lattice descending through G_Γ ;
2. the target choice $\mathcal{P}_\Gamma$ of maximal mutually local magnetic/dyonic line spectrum at a reference theta class;
3. the magnetic and dyonic lattice satisfying mutual locality on that branch;
4. the screening subgroups generated by dynamical electric and magnetic matter;
5. the theta action that transports the branch through its Witten-effect orbit.

We package them as

$$\mathcal{L}_\Gamma^P = (\mathcal{P}_\Gamma, \Lambda_{e,\Gamma}^{\text{adm}}, \Lambda_{m,\Gamma;\mathcal{P}_\Gamma}^{\text{adm}}, \Lambda_{\text{dy},\Gamma;\mathcal{P}_\Gamma}^{\text{ml}}, \mathcal{S}_\Gamma, \rho_\Theta).$$

Here ρ_Θ is the Witten-effect action of the theta identification lattice.

Definition 11.1 (Line-equivalence normal form). Two presentations of the line sector are equivalent when there is an isomorphism of their selected maximal mutually local branches and electric/magnetic charge lattices preserving the DSZ pairing, screening subgroups, theta monodromy, and identification of local dynamical charges.

Theorem 11.2 (Genuine-line reconstruction at fixed branch, SYM9-Q07). *Fix G_Γ , $\mathcal{O}_{\text{line}}$, the dynamical matter content, magnetic policy, and line-spectrum branch $\mathcal{P}_\Gamma$ at a reference theta class. The electric descent lattice, the selected DSZ-compatible magnetic/dyonic extension, screening data, and Witten-effect orbit determine the charge-level line-sector normal form $\mathcal{L}_\Gamma^P$ up to the equivalence of theorem 11.1. A change in basis, center generator, Abelian charge unit, theta-lattice basis, or Dirac-string chart that preserves these structures is a lawful redescription. A change in Γ , $\mathcal{P}_\Gamma$, the DSZ pairing, screening subgroup, or theta monodromy changes the completed line target.*

Proof. Electric descent fixes the globally defined character classes. The target coordinate $\mathcal{P}_\Gamma$ fixes which maximal mutually local magnetic/dyonic extension is admitted at the reference theta class; DSZ compatibility alone is not promoted into a uniqueness theorem for that choice [1]. Screening identifies the charges that can end on dynamical operators, and the theta lattice transports the selected branch through its Witten-effect orbit. These are precisely the charge data entering the action and fusion of invertible line probes at the present scope. Any coordinate change preserving the branch, lattices, pairing, and distinguished screening subgroups induces an isomorphism of the line category at the charge level. Conversely, changing one of those invariant structures changes either the line objects, their braiding/Dirac pairing, their ability to end, or their theta transport, and hence changes the extended operator target. $\square$

11.2 Charge-level theorem and categorical nonclaim

The theorem is a charge-lattice normal form on a fixed branch. It does not claim that the gauge group alone selects a unique maximal mutually local line spectrum, reconstruct every non-topological dynamical property of every line, determine every representation-theoretic multiplicity, or supply the full higher category of junctions. Alternative consistent branches or discrete theta-like parameters, where

present, are additional target coordinates rather than counterexamples to the fixed-branch theorem. The 2026 classification of Alonso et al. constructs Standard Model dyonic lattices and their theta orbits directly; its branch-resolved results are compatible with the present P-indexed formulation and provide an independent comparison at the common charge-lattice scope [13].

11.3 Global-form comparison

For the explicit reference branch $\mathcal{P}_{\Gamma,T}$, the four global forms differ monotonically at the displayed charge level:

- increasing $|\Gamma|$ restricts the electric descent lattice;
- it enlarges the available fractional magnetic lattice on the selected branch;
- it reduces the residual electric one-form symmetry;
- it changes the theta identification lattice.

No one of these fields alone defines the global form. Their joint consistency at a fully specified target is the relevant invariant statement.

Physics translation

The global quotient is not an extra group name pasted onto one fixed theory. It constrains the electric side of a global line problem. A completed line target additionally fixes the maximal mutually local branch, screening, magnetic policy, and theta orbit. Once those fields are declared, the resulting line packet is rigid under lawful presentation changes.

III

Generalized and topological structures

12 One-form symmetry reconstructed from line and monopole data

12.1 Generalized symmetry acts on lines

A one-form global symmetry acts on line operators rather than point-local operators. Its charged objects are classified by the unscreened line-charge group, and its topological symmetry operators have codimension two in four spacetime dimensions [3]. The relevant Standard Model structure has electric and magnetic factors with different origins.

The electric factor is finite and arises from the common local-matter kernel left unquotiented by Γ . The magnetic factor is continuous when the hypercharge Bianchi identity is unbroken by dynamical monopoles.

Theorem 12.1 (One-form reconstruction, SYM9-Q08). *Fix the renormalizable Standard Model target $D_{\text{SM},\Gamma}^{\text{line}}$ with no dynamical hypercharge monopoles. Then*

$$\mathcal{A}_{\Gamma}^{(1)} \cong \text{U}(1)_{\text{mag}}^{(1)} \times (\mathbb{Z}_6/\Gamma)_{\text{el}}^{(1)}.$$

Writing $n = |\Gamma|$, the magnetic factor is generated by the conserved two-form current

$$J_{\Gamma}^{(2)} = \star \frac{n \, da_Y}{2\pi}, \quad d \star J_{\Gamma}^{(2)} = 0,$$

normalized so that the minimally allowed magnetic probe has unit charge. The electric factor acts on Wilson-line classes and is $\mathbb{Z}_{6/n}$.

If a lattice Λ_{mon} of dynamical monopole charges is admitted, the magnetic factor is reduced to the subgroup of $U(1)_{\text{mag}}^{(1)}$ whose characters are trivial on Λ_{mon} . A unit normalized dynamical monopole breaks the continuous magnetic factor completely.

Proof. The electric statement is [theorem 9.1](#). For the magnetic statement, the hypercharge field strength obeys the Bianchi identity $dF_Y = 0$ in the absence of magnetic sources. Hence the dual two-form current is conserved. In the quotient by $\mathbb{Z}_n$, the allowed magnetic flux can be fractional in units $1/n$; multiplying by n yields an integer-normalized one-form charge, giving [theorem 12.1](#) [12]. Electric and magnetic factors commute at the global-symmetry-group level before additional background-action data are chosen, producing the product [theorem 12.1](#). Dynamical monopoles provide endpoints for magnetic lines and explicitly violate the Bianchi conservation by delta-function sources, leaving only the annihilator of their charge lattice. $\square$

12.2 Table by global form

Table 2: One-form symmetry in the no-dynamical-monopole target. The discrete magnetic subgroup dual to the quotient is contained in the displayed continuous magnetic factor.

Γ	n	Electric factor	Full one-form group
1	1	$\mathbb{Z}_6$	$U(1)_{\text{mag}}^{(1)} \times \mathbb{Z}_6^{(1)}$
$\mathbb{Z}_2$	2	$\mathbb{Z}_3$	$U(1)_{\text{mag}}^{(1)} \times \mathbb{Z}_3^{(1)}$
$\mathbb{Z}_3$	3	$\mathbb{Z}_2$	$U(1)_{\text{mag}}^{(1)} \times \mathbb{Z}_2^{(1)}$
$\mathbb{Z}_6$	6	1	$U(1)_{\text{mag}}^{(1)}$

12.3 Why the group is target dependent

The result changes under three common target updates.

1. New electric matter charged under $\mathbb{Z}_6/\Gamma$ screens the electric factor.
2. Dynamical monopoles break the magnetic factor.
3. A different operator category can omit the line operators on which the one-form group acts, making the action unavailable rather than false.

Thus “the Standard Model has a one-form symmetry” is incomplete without Γ , matter content, monopole policy, and operator category.

12.4 Fractionalization and mixed background response

The abstract group [theorem 12.1](#) does not by itself specify how every lower-form symmetry acts in the presence of its electric one-form factor. Throughout this interface we use the Hsin–Gomis $U(1)_{B-L}$ convention. Let B_q denote their quark-number generator. Then

$$Q_{B-L}^{\text{HG}} = \frac{1}{3}B_q - \sum_{i=e,\mu,\tau} L_i. \quad (2)$$

Equivalently, with conventional baryon number $B_{\text{conv}} = B_q/3$, one has $Q_{B-L}^{\text{HG}} = B_{\text{conv}} - \sum_i L_i$. Thus quark fields carry projective charges in thirds, leptons carry integral charges, and minimally charged gauge-invariant local operators have unit charge. The numerical response coefficient below belongs to this fixed zero-form charge convention and is not invariant under an untransported rescaling of Q_{B-L}^{HG} .

For this $U(1)_{B-L}$ background, Hsin and Gomis show that the action requires a fractionalization class

$$k \in H^2(BU(1), \mathbb{Z}_6/\Gamma) \cong \mathbb{Z}_6/\Gamma.$$

Writing $\Gamma = \mathbb{Z}_n$, the corresponding fractional topological transport coefficient is

$$\xi_{\text{HG}}(\Gamma, k) = \frac{n}{3} + \frac{kn}{6} \pmod{1}. \quad (3)$$

At this fixed charge normalization, the same background fields with different k need not define the same symmetry action or response packet. In the target fingerprint, this information is carried by F ; in the global packet it is carried by $\mathcal{F}_\Gamma$. The formula is used as a global-form-sensitive discriminant in [section 15](#), not as a claim that $B - L$ has been promoted to a gauge symmetry.

Electric and magnetic one-form symmetries of an Abelian gauge sector can also carry a mixed ‘t Hooft anomaly. The one-form theorem fixes the symmetry groups; anomaly, fractionalization, charge normalization, and response data are additional fields of the completed packet rather than alternative definitions of those groups [12].

Adversarial check

A quotient by Γ does not by itself imply an exact magnetic symmetry, and a background-field list does not by itself specify a lower-form action when fractionalization is available. The global form, monopole policy, line branch, and applicable fractionalization class are independent target coordinates whose combination determines the completed result.

13 Theta identifications and physical phase quotients

13.1 Gauge-topological theta-identification torus

The local action admits gauge-topological coordinates

$$(\theta_2, \theta_3, \tilde{\theta}) \in \mathbb{R}^3,$$

where $\tilde{\theta}$ is written in integer hypercharge normalization. At fixed matter-phase convention and on the selected line-spectrum branch $\mathcal{P}_{\Gamma, T}$, the global bundle data and Witten effect identify coupled shifts. The resulting gauge-topological, fixed-line-branch torus is

$$\Theta_\Gamma^{\text{line}} = \mathbb{R}^3 / \Lambda_{\theta, \Gamma, \mathcal{P}_{\Gamma, T}}. \quad (4)$$

This quotient records when theta-coordinate representatives return the declared gauge-topological line packet to the same Witten-effect orbit. It is not, by itself, the complete physical phase space of the renormalizable Standard Model, because anomalous fermion rephasings act simultaneously on theta coordinates and on Yukawa or mass phases.

A convenient basis of the identification lattice, in the coordinate order $(\theta_2, \theta_3, \tilde{\theta})$, is:

$$\begin{aligned} \Gamma = 1 : & \quad (2\pi, 0, 0), (0, 2\pi, 0), (0, 0, 2\pi), \\ \Gamma = \mathbb{Z}_2 : & \quad (2\pi, 0, 4\pi), (0, 2\pi, 0), (0, 0, 8\pi), \\ \Gamma = \mathbb{Z}_3 : & \quad (2\pi, 0, 0), (0, 2\pi, 6\pi), (0, 0, 18\pi), \\ \Gamma = \mathbb{Z}_6 : & \quad (2\pi, 0, 36\pi), (0, 2\pi, 48\pi), (0, 0, 72\pi). \end{aligned} \quad (5)$$

These bases encode the fact that shifting a non-Abelian theta coordinate can require a compensating hypercharge-theta shift to return the selected line spectrum to the same Witten-effect orbit [2].

Theorem 13.1 (Theta-identification and physical-phase theorem, SYM9-Q09). *For the four global-form targets on the branch $\mathcal{P}_{\Gamma, T}$ at line-operator scope, the gauge-topological theta-identification torus is $\Theta_\Gamma^{\text{line}}$ with lattice bases given in [equation \(5\)](#). The covolume of $\Lambda_{\theta, \mathbb{Z}_n, \mathcal{P}_{\Gamma, T}}$ relative to the naive $(2\pi\mathbb{Z})^3$ lattice is n^2 , and the hypercharge coordinate has range $2\pi n^2$ in the displayed fundamental domain.*

A shift by a lattice vector preserves the gauge-topological line packet on the declared branch. A shift outside the lattice changes the fixed-matter-phase-convention Witten-effect line presentation unless it is accompanied by an admitted anomalous fermion rephasing and the corresponding transport of Yukawa or mass phase data. The full physical phase object is therefore the anomalous-rephasing quotient defined below, not $\Theta_\Gamma^{\text{line}}$ alone.

Proof. The quotient permits hypercharge fluxes in units $1/n$, so the Abelian instanton-density integral can occur in units $1/n^2$. Its conjugate theta coordinate therefore has period $2\pi n^2$. For $\Gamma = \mathbb{Z}_2$ and $\mathbb{Z}_3$, rewriting the topological action in $U(2)$ and $U(3)$ variables gives the coupled shifts displayed above. For $\mathbb{Z}_6$, compatibility with both quotient factors yields the 36π and 48π compensating shifts and the 72π pure Abelian period. Equivalently, the Witten effect maps the selected dyonic line branch to itself under precisely these shifts. The determinants of the basis matrices in units of 2π are 1, 4, 9, 36, proving the covolume statement.

These calculations determine the gauge-topological line torus at a fixed matter-phase convention. An anomalous change of fermion variables can shift the theta coordinates while transporting phases in the mass and Yukawa tensors. Such a transformation acts on the product of the line torus and the matter-phase data, which establishes the final distinction in the theorem. $\square$

13.2 Anomalous fermion rephasings and the physical phase packet

Let $\mathcal{Y}_{\text{ph}}$ denote the mass- and Yukawa-phase data retained by the frozen target, and let $\mathcal{R}_{\text{anom}}$ denote the admitted anomalous fermion rephasings acting simultaneously on $\Theta_\Gamma^{\text{line}}$ and $\mathcal{Y}_{\text{ph}}$. The complete physical phase record is

$$\Phi_\Gamma^{\text{phys}} = (\Theta_\Gamma^{\text{line}} \times \mathcal{Y}_{\text{ph}}) / \mathcal{R}_{\text{anom}}. \quad (6)$$

This quotient separates a gauge-topological line-periodicity statement from a statement about physically inequivalent Standard Model phase data. Alonso et al. independently classify the Standard Model dyonic lattices and theta domains and show that quotienting the anomalous $B + L$ direction reduces the gauge-theta three-torus to a physical two-torus; the present $\Theta_\Gamma^{\text{line}} / \Phi_\Gamma^{\text{phys}}$ distinction is the target-typed form of that separation [13].

For baryon- or lepton-number rephasings in the renormalizable target,

$$\theta_2 \mapsto \theta_2 - \alpha, \quad \tilde{\theta} \mapsto \tilde{\theta} + 18\alpha.$$

The invariant electroweak combination is therefore

$$\vartheta_{\text{EW}} = \tilde{\theta} + 18\theta_2. \quad (7)$$

The familiar statement that the weak theta coordinate can be shifted away is thus a statement about a lawful anomalous rephasing. When magnetic or boundary probes render the Abelian theta term detectable, the invariant combination [equation \(7\)](#), rather than either coordinate separately, enters the physical phase packet.

In the strong sector, θ_3 combines with the quark-mass determinant into the imported $\bar{\theta}$ class. The present paper does not rederive that quotient or its phenomenological smallness. It records how the global-form line torus coexists with the separately established strong-CP and Yukawa/mixing phase discipline [30, 33, 7].

13.3 CP/T-fixed loci of the gauge-topological theta sector

The CP or time-reversal involution on $\Theta_\Gamma^{\text{line}}$ sends the theta vector to its negative. A fixed class is a solution of

$$2\vartheta \in \Lambda_{\theta, \Gamma, \mathcal{P}_{\Gamma, T}}.$$

In convenient fundamental domains, the possibilities are [2]:

- $\Gamma = 1$: each coordinate is 0 or π ;
- $\Gamma = \mathbb{Z}_2$: $\theta_3 = 0, \pi$; if $\theta_2 = 0$, then $\tilde{\theta} = 0, 4\pi$, while if $\theta_2 = \pi$, then $\tilde{\theta} = 2\pi, 6\pi$;
- $\Gamma = \mathbb{Z}_3$: $\theta_2 = 0, \pi$; if $\theta_3 = 0$, then $\tilde{\theta} = 0, 9\pi$, while if $\theta_3 = \pi$, then $\tilde{\theta} = 3\pi, 12\pi$;
- $\Gamma = \mathbb{Z}_6$: the eight fixed points are

θ_2	θ_3	$\tilde{\theta}$
0	0	0, 36 π
0	π	12 π , 48 π
π	0	18 π , 54 π
π	π	30 π , 66 π .

These are coordinate representatives of fixed classes in the gauge-topological theta torus, not primitive topological identities. They are not sufficient conditions for CP or time-reversal invariance of the full Standard Model: the complete condition is imposed on $\Phi_\Gamma^{\text{phys}}$, including the Yukawa/mixing and strong-sector phase data governed by SYM-4, SYM-5, and the strong-CP interface.

Physics translation

A theta coordinate is a representative. At fixed global form and line branch, $\Theta_\Gamma^{\text{line}}$ is the gauge-topological line-identification torus. The physical phase object is $\Phi_\Gamma^{\text{phys}}$, obtained only after the matter-phase data and admitted anomalous rephasings are included. A lawful anomalous rephasing can move a representative in $\Theta_\Gamma^{\text{line}}$ while preserving $\Phi_\Gamma^{\text{phys}}$ when the mass and Yukawa phases are transported with it. Changing the line branch, the theta lattice, or the physical phase class changes the completed packet.

14 Electroweak fixation: the line and monopole interface

14.1 Import from SYM-8

SYM-8 proves that the one-Higgs-doublet fixation leaves the connected generator

$$Q = T_3 + Y = T_3 + \frac{\tilde{Y}}{6}$$

and one massless electromagnetic field direction [9]. SYM-9 now asks how the pre-fixation electric and magnetic line labels project to the residual electromagnetic sector.

For an electric line with weak weight λ_2^e and integer hypercharge q ,

$$Q_{\text{em}} = \frac{q}{6} + \lambda_2^e.$$

A magnetic line remains deconfined through the Higgs phase only when its weak and hypercharge magnetic charges satisfy the Higgs compatibility condition; in the normalization of [2],

$$6g = \lambda_2^m, \quad G_{\text{em}} = 6g.$$

Other magnetic lines acquire an area law in the selected Higgs phase.

14.2 Minimum probe charges

The resulting minimum electromagnetic probe charges depend on Γ :

For $\Gamma = \mathbb{Z}_3$ and $\mathbb{Z}_6$, the naive product QG can be fractional if color magnetic charge is ignored. The full DSZ pairing includes the $SU(3)$ contribution and remains integral. Equivalently, the low-energy global group relevant to the combined color/electromagnetic sector is $U(3)$ rather than a direct product $SU(3) \times U(1)$ [2].

Table 3: Minimum electromagnetic electric and magnetic probe charges after one-doublet electroweak fixation, with the electron normalized to $Q = -1$. These are line-sector minima, not necessarily dynamical particle charges.

Γ	$ Q _{\min}$	$ G _{\min}$	Magnetic qualification
1	1/6	6	Pure electromagnetic description is sufficient.
$\mathbb{Z}_2$	1/3	3	Pure electromagnetic description is sufficient.
$\mathbb{Z}_3$	1/6	2	Minimal monopole carries color magnetic charge.
$\mathbb{Z}_6$	1/3	1	Minimal monopole carries color magnetic charge.

14.3 Electromagnetic theta

Projecting the topological terms through electroweak fixation gives

$$\theta_{\text{em}} = \frac{\tilde{\theta} + 18\theta_2}{4} = \frac{\vartheta_{\text{EW}}}{4}.$$

The line-sensitive identification range of θ_{em} depends on Γ and the minimum charge lattice. The coordinate is the electroweak projection of the anomaly-invariant combination in $\Phi_{\Gamma}^{\text{phys}}$, not an independently physical theta angle. In particular, the maximal quotient permits a conventional 2π electromagnetic range despite fractional quark electric charges because the color and electromagnetic global structures are correlated.

14.4 No conflict with residual-gauge classification

SYM-8 distinguished the residual local gauge redundancy from global/asymptotic electromagnetic charge action. The present line-sector analysis is compatible with that distinction. Wilson and 't Hooft probes are extended operators in the residual gauge theory; they do not turn proper local gauge transformations into ordinary global symmetries. They reveal the global completion of the gauge architecture.

Exact nonclaim boundary

The table does not predict a dynamical monopole or fractionally charged particle. It classifies the minimum charges of admissible probes. Turning a probe into a dynamical state requires a separate ultraviolet or completeness premise.

15 Distinct global packets are distinct completed targets

15.1 Extended equivalence

At local scope, two theories are compared through their local observable algebras and perturbative sectors. At line-completed scope, an equivalence must also preserve the line objects, their fusion and endability, DSZ pairing, one-form charges, gauge-topological theta monodromy, physical phase classes, and admitted background action and response.

Definition 15.1 (Extended-target equivalence). Two completed targets D_1 and D_2 are equivalent at the present scope if there is an isomorphism preserving:

1. the local field theory and local observable algebra;

2. the selected maximal mutually local line branch, genuine line-charge lattice, and DSZ pairing;
3. electric and magnetic screening subgroups;
4. one-form symmetry action;
5. gauge-topological theta identification and Witten-effect transport;
6. the physical phase quotient under admitted anomalous fermion rephasings;
7. every admitted background transformation law, fractionalization or extension class, topological response, response-counterterm class, and defect action.

Theorem 15.2 (Global-form separation, SYM9-Q10). *Let D_1 and D_2 be two admitted completed targets. If their packets differ in a genuine line class, selected maximal mutually local branch, screening quotient, magnetic charge class, one-form action, gauge-topological theta identification, physical phase class, fractionalization/background-action class, admitted topological response, or defect action, then*

$$D_1 \neq D_2$$

at the declared extended-operator scope. Equality of the local Lie algebra and local perturbative correlators does not restore equivalence.

Proof. An equivalence at extended scope must induce a bijection on admitted extended operators and preserve their observable relations. If a line exists in one target and not the other, or if corresponding lines have different DSZ pairing or endability, no such bijection exists. The same is true if the one-form action, theta monodromy, physical phase class, symmetry-fractionalization class, or background response differs, because these are encoded in linking phases, junction laws, and correlation functions involving the admitted extended operators and backgrounds. Therefore any listed packet difference obstructs extended-target equivalence. Local agreement is insufficient because it forgets the very objects on which the obstruction is measured. $\square$

15.2 Fractionalization-sensitive background response

For the $U(1)_{B-L}$ background-response target of theorem 2.6, write $\Gamma = \mathbb{Z}_n$ and

$$k \in H^2(BU(1), \mathbb{Z}_6/\Gamma) \cong \mathbb{Z}_{6/n}.$$

The response coefficient equation (3) depends on both n and k and is quoted in the Hsin–Gomis $B - L$ convention fixed in section 12.4. Consequently, equality of the background-field set does not imply equality of the background symmetry action: different k can change the junction decoration, mixed response, and anomaly data even at fixed Γ . The completed packet therefore records $\mathcal{F}_T$, not merely the names of the background fields. This is the precise reason F is part of the target fingerprint.

15.3 Examples of separating data

The following are independent practical discriminants.

1. The electric one-form group is $\mathbb{Z}_6$, $\mathbb{Z}_3$, $\mathbb{Z}_2$, or trivial.
2. The minimum electromagnetic magnetic probe charge is 6, 3, 2, or 1 after electroweak fixation.
3. The hypercharge theta range on the declared line branch is 2π , 8π , 18π , or 72π in integer normalization.
4. The fractional topological transport coefficient involving $B - L$ and the magnetic one-form current distinguishes combinations of global form and fractionalization class [12].

Any one of these, when included in the target and measured as an invariant, is enough to block the claim that the two completed theories are merely different notations.

15.4 Sharp-local and defect-completed identity

There is no contradiction in saying both

$$D_{\text{SM},\Gamma_1}^{\text{loc}} \simeq D_{\text{SM},\Gamma_2}^{\text{loc}}$$

and

$$D_{\text{SM},\Gamma_1}^{\text{def}} \neq D_{\text{SM},\Gamma_2}^{\text{def}}.$$

The equivalence relation is operator-category indexed. SYM-11 must preserve that index, together with any line-branch and fractionalization coordinates, when stating closure.

15.5 No metaphysical promotion

Calling the global form target-bearing does not assert that the group manifold is an additional material substance. It means that changing the quotient changes the admissible global bundles and extended observables. Calling a fractionalization class target-bearing at background-probe scope likewise means that it changes the admitted symmetry action and response; it does not promote a cocycle representative into an independent physical substance. The structural status follows from a variation test, not from reverence for notation.

16 Higher-flavor symmetry as a target-indexed 2-group interface

16.1 Why ordinary direct-product language can fail

A zero-form flavor symmetry and a one-form magnetic symmetry need not combine as a direct product. Mixed flavor–flavor–hypercharge triangle coefficients can deform the background transformation law so that the two symmetries form a 2-group. The charges may still resemble those of a product, but the current algebra, Ward identities, and background gauge transformations are intertwined [14, 31].

Consider an Abelian flavor background $A^{(1)}$ and a two-form background $B^{(2)}$ for a magnetic one-form symmetry. In a fixed normalization, a representative 2-group law is

$$\begin{aligned} A^{(1)} &\mapsto A^{(1)} + d\lambda^{(0)}, \\ B^{(2)} &\mapsto B^{(2)} + d\Lambda^{(1)} + \frac{\widehat{\kappa}}{2\pi} \lambda^{(0)} F_A^{(2)}. \end{aligned} \tag{8}$$

with a quantized structure coordinate $\widehat{\kappa}$ relative to the chosen current, background, and period lattice. The corresponding gauge-invariant three-form field strength is modified by a Chern–Simons term.

16.2 The normalization-covariant extension datum

The bare number $\widehat{\kappa}$ is not meaningful without its current and background normalization. We therefore use the extension datum

$$\mathcal{K}_{F,\Gamma} = [J^{(2)}, B^{(2)}, \text{Per}(B^{(2)}), \widehat{\kappa}_F]_{\text{norm}}. \tag{9}$$

where the equivalence relation transports the coupled quantities together. If

$$J'^{(2)} = rJ^{(2)}, \quad B'^{(2)} = r^{-1}B^{(2)},$$

so that the background coupling is unchanged, then

$$\text{Per}(B'^{(2)}) = r^{-1} \text{Per}(B^{(2)}), \quad \widehat{\kappa}'_F = r^{-1} \widehat{\kappa}_F. \tag{10}$$

Thus the background law and its period lattice are invariant data; the displayed coefficient is a coordinate on that datum.

Córdova and Koren use an integer hypercharge convention and write their source-coordinate structure constant as

$$\widehat{\kappa}_F^{\text{CK}} = -\frac{\kappa_F^{\text{CK}}}{2}. \quad (11)$$

This source relation is quoted only in that current/background convention [15].

The one-form theorem in this manuscript instead uses the quotient-unit current

$$J_\Gamma^{(2)} = n \star \frac{da_Y}{2\pi}, \quad n = |\Gamma|.$$

Let r_Γ denote the conversion defined by

$$J_\Gamma^{(2)} = r_\Gamma J_{\text{CK}}^{(2)}.$$

Then the corresponding quotient-coordinate coefficient is

$$\widehat{\kappa}_{F,\Gamma} = r_\Gamma^{-1} \widehat{\kappa}_F^{\text{CK}},$$

with the two-form background and its period lattice transported as in equation (10). If the Córdova–Koren magnetic current is identified with the unquotiented current $\star da_Y/(2\pi)$, then $r_\Gamma = n$ and

$$\widehat{\kappa}_{F,\Gamma} = -\frac{\kappa_F^{\text{CK}}}{2n}$$

in the quotient-unit coordinate. The theorem below does not require this optional coordinate identification; it requires the normalization class equation (9).

16.3 Standard Model flavor application

In the zero-Yukawa limit, the kinetic and gauge terms admit the flavor group conventionally summarized as $U(3)^5$, subject to the Abelian and anomaly qualifications fixed in SYM-5. Triangle diagrams with two flavor currents and the hypercharge current have quantized coefficients κ_F . When $\kappa_F \neq 0$, the flavor symmetry and magnetic hypercharge one-form symmetry form a higher group rather than an independent direct product [15].

The essential operator relation is schematically

$$\partial^\mu J_\mu(x) J_\nu(y) \sim \kappa_F \partial^\lambda \delta(x-y) \epsilon_{\lambda\nu\rho\sigma} F_Y^{\rho\sigma}(y),$$

which shows that the zero-form currents do not close independently in the presence of the one-form current.

Theorem 16.1 (Higher-group interface with normalization transport, SYM9-Q11). *Fix the zero-Yukawa Standard Model target $D_{Y=0}^{\text{bg}}$, an exact admitted flavor subgroup F , background fields for F and the magnetic hypercharge one-form symmetry, a global form Γ , and a magnetic-current/two-form-background normalization class $\mathcal{K}_{F,\Gamma}$. Let κ_F be the corresponding nonzero quantized flavor–flavor–hypercharge triangle coefficient in a declared source convention. Then the background transformations and current algebra realize a nontrivial 2-group*

$$F^{(0)} \times_{\mathcal{K}_{F,\Gamma}} U(1)_{Y,\text{mag}}^{(1)}.$$

In the Córdova–Koren source convention, the coefficient coordinate is $\widehat{\kappa}_F^{\text{CK}} = -\kappa_F^{\text{CK}}/2$. In any quotient-normalized convention, its numerical coordinate is obtained only by the simultaneous current, background, and period-lattice transport of equation (10); no Γ -independent equality for the bare coefficient is asserted across those conventions.

If ultraviolet monopoles break the magnetic one-form symmetry, an unbroken ultraviolet flavor symmetry must lie in a combination for which the relevant higher-group obstruction is canceled or modified by additional ultraviolet matter. The theorem does not assign the full zero-Yukawa 2-group as an exact symmetry of the generic nonzero-Yukawa Standard Model.

Proof. The nonzero quantized triangle coefficient produces the modified current algebra and background transformation law [equation \(8\)](#); these are the defining data of the continuous 2-group. Quantization prevents the extension class from being removed by a continuous symmetry-preserving deformation. Equations [equations \(9\) and \(10\)](#) show that rescaling the magnetic current changes the coordinate value of the coefficient unless the two-form background and its period lattice are transported contragrediently. Hence the source relation [equation \(11\)](#) and the quotient-unit current of SYM9-Q08 define the same physical extension only through the declared normalization class.

If ultraviolet monopoles break the magnetic one-form symmetry, the higher-group relation forbids retaining the zero-form factor unchanged unless the coefficient is canceled or the symmetry is explicitly broken; additional massive matter can supply the required coefficient shift. Nonzero Standard Model Yukawa matrices explicitly break the independent flavor factors and therefore move the exact 2-group statement to the stated limit target. $\square$

16.4 Yukawa spurions

The Yukawa matrices can be treated as spurions organizing the breaking of the zero-form flavor group and the associated higher-group structure. This bookkeeping is useful, but a spurion transformation does not restore an exact physical symmetry at fixed numerical Yukawas. The exact target remains the zero-Yukawa limit; the physical Standard Model carries controlled breaking data.

16.5 Global-form dependence

The magnetic hypercharge one-form symmetry is normalized by the global quotient and can be broken by monopoles. Therefore the higher-group record must carry Γ , magnetic policy, the current/background normalization, and the two-form period lattice even when the local triangle coefficient is unchanged. A higher-flavor statement without those fields is incomplete.

Adversarial check

A nonzero triangle coefficient establishes the higher-group extension only together with the current and background convention in which its numerical coefficient is quoted. Copying $\widehat{\kappa}_F^{\text{CK}} = -\kappa_F^{\text{CK}}/2$ unchanged into the quotient-unit current coordinate is not a lawful specialization. Nor does the exact zero-Yukawa result imply that the observed nonzero-Yukawa Standard Model has an exact $U(3)^5$ higher symmetry.

17 The noninvertible frontier: massless QED and QCD fibers

17.1 From anomalous axial rotations to topological defects

In massless QED the classical axial $U(1)_A$ current is broken by the Adler–Bell–Jackiw anomaly. Nevertheless, for every rational angle $2\pi p/N$, an axial rotation can be combined with a suitable three-dimensional topological quantum field theory coupled to the electromagnetic gauge field to form a conserved gauge-invariant codimension-one topological operator $\mathcal{D}_N^p$ [11].

These operators do not obey an ordinary group law. Their fusion involves sums or condensation defects with TQFT coefficients. They act as the expected axial rotation on local fermion operators and Wilson lines, but their action on 't Hooft lines can attach fractional surfaces or annihilate states on nontrivial cycles. This is the operational noninvertibility.

Theorem 17.1 (Noninvertible target-fiber theorem, SYM9-Q12). *The operators $\mathcal{D}_N^p$ of [11] define exact noninvertible generalized symmetries on the target fibers*

$$D_{\text{QED}}^{m_e=0, \text{def}} \quad \text{and} \quad D_{\text{QCD}}^{m_u=m_d=0, \text{IR}}$$

with the required electromagnetic line and defect categories. In QED they act invertibly as rational axial rotations

on local fermions and Wilson lines but noninvertibly on 't Hooft-line sectors. In first-generation massless QCD below the electroweak scale, matching the defect requires the $\pi^0 F \wedge F$ coupling in the pion effective theory.

Nonzero fermion masses explicitly break the axial action and remove the exact target premise. Therefore these defects are not assigned as exact symmetries of the generic massive Standard Model without a separate deformation theorem.

Proof. The cited construction stacks the rational axial rotation with a fractional quantum Hall TQFT so that the gauge variation induced by the ABJ anomaly is canceled on the defect. The resulting operator is topological and gauge invariant. Its fusion with its orientation reverse produces a condensation operator rather than the identity, and its action on 't Hooft lines is not bijective, establishing noninvertibility. In the QCD target the same symmetry must be matched across the renormalization-group flow; the pion coupling supplies the required infrared response. Mass terms are not invariant under the axial rotation and therefore violate the topological conservation law, so exactness is limited to the stated massless fibers. $\square$

17.2 Relation to ordinary anomaly language

The noninvertible construction does not erase the ABJ anomaly. It reorganizes the surviving topological content after the continuous invertible axial group is lost. The anomaly is part of the construction data and matching condition.

17.3 Relation to the four global forms

The particular massless QED/QCD defects are not a replacement for the global-form line classification. They require electromagnetic 't Hooft lines and can depend on the available one-form structures, but their exact fusion category is additional defect data. Other noninvertible structures can be produced by higher gauging finite one-form subgroups, including structures associated with the Standard Model center. Those broader completions are not claimed here because their full fusion and target requirements lie beyond the frozen theorem.

17.4 No generic-core promotion

Exact nonclaim boundary

The physical electron and light-quark masses are nonzero. The theorem therefore records exact noninvertible symmetries in controlled massless targets and their matching significance; it does not describe them as exact symmetries of the measured massive Standard Model. Approximate or softly broken remnants would require a separate quantitative analysis.

17.5 Why the result belongs in SYM-9

These defects act on extended operators and their fusion is not a group. They therefore test whether the symmetry-status calculus can accommodate structures beyond ordinary zero-form Lie groups. SYM-9's contribution is to place them in the correct target fiber and prevent either omission or overgeneralization.

IV

Structural integration, normal form, and handoff

18 Global-completion covariance and the AASC role map

18.1 Lawful presentation changes

The completed packet is displayed through choices: a local trivialization, a center generator, a hypercharge unit, bases for electric and magnetic lattices, a representative of the fixed line-spectrum branch, a theta fundamental domain, a fermion-phase convention, a Dirac-string chart, background gauges, cocycle representatives for fractionalization or extension data, and defect representatives. These choices are not themselves new global targets.

Definition 18.1 (Lawful global-completion redescription). A lawful redescription is an invertible transformation that preserves the complete target fingerprint and induces isomorphisms on:

1. G_Γ as an embedded global-form target;
2. the selected maximal mutually local line branch, electric character lattice, and magnetic cocharacter lattice;
3. the DSZ pairing and screening subgroups;
4. the one-form action, gauge-topological theta monodromy, and physical phase quotient;
5. all admitted background transformation laws, current/background normalizations and period lattices, fractionalization/extension classes, topological responses, and defect fusion/action data.

Examples include local gauge/trivialization changes, replacement of ξ by another generator of the same subgroup, simultaneous Abelian normalization transport, integral basis changes preserving the pairing, theta-lattice basis changes, anomalous fermion rephasings with the induced transport of theta and mass/Yukawa phases, coboundary changes representing the same fractionalization class, and equivalent presentations of the same background or defect.

Theorem 18.2 (Global-completion covariance, SYM9-Q13). *If two presentations are related by a lawful redescription in the sense of theorem 18.1, then they determine the same global-completion packet up to canonical isomorphism. In particular, the global form, line-spectrum branch, genuine line-charge normal form, screening quotient, one-form symmetry, gauge-topological theta torus, physical phase class, fractionalization/background-action class, higher-group normalization class $\mathcal{K}_{F,\Gamma}$, and noninvertible fusion/action class are invariant.*

Proof. Each listed transformation is, by definition, an isomorphism of the typed structures entering the packet. Local gauge and trivialization changes preserve the global bundle isomorphism class. A center-generator change is an automorphism of the same subgroup. Normalization and integral basis changes transport dual lattices and pairings contragrediently. A theta-lattice basis change preserves the gauge-topological torus and monodromy. An admitted anomalous fermion rephasing acts simultaneously on the theta coordinates and mass/Yukawa phases and therefore preserves the quotient $\Phi_\Gamma^{\text{phys}}$. A rescaling of a one-form current is lawful only with the contragredient background, coefficient, and period-lattice transport; it therefore preserves $\mathcal{K}_{F,\Gamma}$ while changing the displayed coefficient coordinate. A coboundary or equivalent junction decoration preserves the cohomology/fractionalization class and its gauge-invariant response. Equivalent background and defect presentations preserve their field strengths, junction laws, and fusion/action classes. Their product therefore induces an isomorphism of the entire packet. $\square$

18.2 Representative versus invariant

18.3 Global-form role theorem

Theorem 18.3 (Global-form role, SYM9-A01). *At the line- or defect-completed target, Γ is a target-bearing structural datum. Varying it changes the admissible global bundles, character/cocharacter constraints, one-form symmetry, and theta-identification package. At sharp-local scope, the distinction may be unavailable because the discriminating operator category has been omitted. The displayed product cover and center generator remain representatives. The theorem does not say that Γ alone determines every line-spectrum branch.*

Proof. The variation consequences were established in theorems 8.1, 10.1, 12.1, 13.1 and 15.2. Since

Table 4: Representative and invariant data in the global-completion problem.

Representative	Invariant content	Blocked promotion
Local gauge/trivialization	Bundle and extended observable classes	Gauge chart does not select Γ .
Hypercharge unit	Character lattice, charge ratios, kernel	Unit choice does not create a new global group.
Generator ξ	Embedded subgroup Γ	Generator notation is not target identity.
Weight/cocharacter basis	Fixed line branch, lattices, and DSZ pairing	Basis vector is not a new line species.
Dirac string	Magnetic charge and linking phase	String placement is not magnetic multiplicity.
Theta fundamental domain	Class in $\Theta_{\Gamma}^{\text{line}} = \mathbb{R}^3 / \Lambda_{\theta, \Gamma, \mathcal{P}_{\Gamma}}$	Coordinate interval is not primitive.
Fermion-phase convention	Class in $\Phi_{\Gamma}^{\text{phys}} = (\Theta_{\Gamma}^{\text{line}} \times \mathcal{Y}_{\text{ph}}) / \mathcal{R}_{\text{anom}}$	Rephasing convention is not a new physical phase.
Background gauge/cocycle representative	Fractionalization or extension class and response	Gauge or cocycle representative is not a new symmetry action.
Magnetic-current normalization	$\mathcal{K}_{F, \Gamma}$ and its background period lattice	A bare $\widehat{\kappa}_F$ is not convention independent.
Defect drawing/TQFT presentation	Fusion and action class	Diagram is not the noninvertible structure itself.

changing Γ changes at least one admitted extended observable, it changes the completed target. The product cover and generator merely present the quotient and can be changed without altering the subgroup action. The independent line-branch coordinate was fixed in [theorem 2.4](#) and is not collapsed into the global-form datum. $\square$

18.4 Generalized-action role theorem

Theorem 18.4 (Generalized-action role, SYM9-A02). *An exact one-form, fractionalized background, higher-group, or noninvertible structure is assigned a physical-action role only on a target whose admitted extended operators, backgrounds, or defects transform nontrivially under it. Its current, linking phase, response coefficient, anomaly, or matching term is a derived witness. A background gauge, cocycle representative, or defect-coordinate choice is presentation skin. A structure requiring a Yukawa, massless, monopole, fractionalization, or defect completion absent from the core is a target fiber or effective route rather than an exact generic-core action.*

Proof. The quotient-action criterion of SYM-0 applies to the appropriate operator category. One-form symmetries act on lines by [theorem 12.1](#); fractionalized lower-form actions change junction and response data by [theorem 2.6](#) and [equation \(3\)](#); the higher-group law acts on background fields and currents by [theorem 16.1](#); and the noninvertible defects act on 't Hooft sectors by [theorem 17.1](#). These are nontrivial physical actions at their targets. The associated response and matching data record the action but do not ground it primitively. Missing target premises redirect the claim rather than permitting an unqualified role assignment. $\square$

18.5 Carrier-support interpretation

The global packet depends on the local gauge and representation architecture without being reducible to it. The support path is

$$\begin{aligned} &\text{transition/transport architecture} \longrightarrow \text{representation and charge lattice} \\ &\quad \longrightarrow \text{common center and global form} \\ &\quad \longrightarrow \text{extended-operator/background/topological packet.} \end{aligned}$$

The support relation is many-to-many. One global form controls several generalized structures, while a complete generalized action can depend additionally on line branch, flavor, anomaly, fractionalization, background, or magnetic data.

19 Global-completion normal form

19.1 Packet definition

For a fixed target fingerprint, define

$$\mathbb{G}_\Gamma = (G_\Gamma, \mathcal{P}_\Gamma, \mathcal{L}_\Gamma^P, \mathcal{S}_\Gamma, \mathcal{M}_\Gamma, \mathcal{A}_\Gamma^{(1)}, \Theta_\Gamma^{\text{line}}, \Phi_\Gamma^{\text{phys}}, \mathcal{F}_\Gamma, \mathcal{H}_\Gamma^{\text{higher}}, \mathcal{N}_\Gamma),$$

where:

- G_Γ is the compact connected global group;
- $\mathcal{P}_\Gamma$ is the chosen maximal mutually local line-spectrum branch at a reference theta class;
- $\mathcal{L}_\Gamma^P$ is the globally admissible mutually local electric/magnetic/dyonic line packet on that branch;
- $\mathcal{S}_\Gamma$ records electric and magnetic endability by dynamical objects;
- $\mathcal{M}_\Gamma$ records the probe and dynamical monopole policy;
- $\mathcal{A}_\Gamma^{(1)}$ is reconstructed from the unscreened line charges and Bianchi current;
- $\Theta_\Gamma^{\text{line}}$ is the branch-indexed gauge-topological theta-identification torus with Witten-effect transport;

- $\Phi_\Gamma^{\text{phys}}$ is the physical phase quotient obtained from $\Theta_\Gamma^{\text{line}}$, the retained mass/Yukawa phases, and the admitted anomalous fermion rephasings;
- $\mathcal{F}_\Gamma$ records admitted background transformation, fractionalization/extension, and response data;
- $\mathcal{H}_\Gamma^{\text{higher}}$ is the target-indexed higher-group packet, including the flavor subgroup and normalization-covariant extension datum $\mathcal{K}_{F,\Gamma}$, if admitted;
- $\mathcal{N}_\Gamma$ is the target-indexed noninvertible defect structure, if admitted.

A null value for $\mathcal{F}_\Gamma$, $\mathcal{H}_\Gamma^{\text{higher}}$, or $\mathcal{N}_\Gamma$ means that the defining target data are absent, not that generalized symmetries have been metaphysically excluded.

Theorem 19.1 (Global-completion normal form, SYM9-A03). *Fix the local Standard Model data, one of the four global forms Γ , the operator category, magnetic policy, line-spectrum branch $\mathcal{P}_\Gamma$, background-field set $\mathcal{B}$, fractionalization or extension datum F , Yukawa/mass and phase data, and spin scope. Then the theorems of theorems 8.1, 9.1, 10.1, 11.2, 12.1, 13.1, 15.2, 16.1 and 17.1 determine a coherent global-completion packet $\mathfrak{G}_\Gamma$ up to lawful redescription. Its components are not independently editable: any proposed change must either be induced by a lawful packet isomorphism, satisfy a declared target update, or break at least one compatibility theorem.*

Proof. G_Γ fixes the electric descent condition. The independent target coordinate $\mathcal{P}_\Gamma$ selects the maximal mutually local magnetic/dyonic extension at a reference theta class; the DSZ pairing then constrains that selected branch rather than being used as a uniqueness theorem for all branches. The dynamical content fixes screening, and these data determine the one-form action. The global bundle and selected branch determine $\Theta_\Gamma^{\text{line}}$ and its Witten-effect orbit. Together with the mass/Yukawa phase data and admitted anomalous fermion rephasings, this determines $\Phi_\Gamma^{\text{phys}}$. The background package $\mathcal{B}$ together with F determines the admitted junction, fractionalization/extension, and response data $\mathcal{F}_\Gamma$. For a higher-group fiber, the triangle coefficient and quotient-normalized one-form current determine the same extension only after the contragredient two-form-background, period-lattice, and coefficient transport of theorem 16.1; the packet stores $\mathcal{K}_{F,\Gamma}$ rather than a bare coefficient. Higher-group and noninvertible entries are then determined, when applicable, by their additional flavor, massless, background, and defect premises. Theorem 18.2 proves invariance under lawful presentation changes. Therefore the packet is coherent and redescription stable at the declared scope. Any non-isomorphic alteration changes a theorem input or output and hence is either a target update or an inconsistency. $\square$

19.2 Family normal form

The completed targets form the controlled family

$$\mathfrak{F}_{\text{SM}}^{\text{def}} = \{ \mathfrak{G}_{\Gamma, \mathcal{P}_\Gamma, F} : \Gamma \in \{1, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_6\}, \mathcal{P}_\Gamma, F \text{ declared when applicable} \},$$

with further fibers for monopole, background, Yukawa, massless, and defect choices. The family is finite in the quotient coordinate and controlled rather than collapsed in the additional target coordinates. The explicit charge and theta tables use the $\mathcal{P}_{\Gamma, \Gamma}$ branch. No claim is made that every alternative discrete line branch has been enumerated for every possible completion.

Theorem 19.2 (Family-indexed closure handoff, SYM9-A04). *Any downstream symmetry closure that admits extended operators must state one of the following scopes:*

1. a sharp-local target $D_{\text{SM}}^{\text{loc}}$ that intentionally forgets global-form discriminants;
2. a fixed completed target $D_{\text{SM}, \Gamma}^{\text{def}}$ with the line branch, monopole policy, background-action/fractionalization data, and all other packet fields declared;
3. an explicitly indexed family of completed targets, with statements quantified over every retained global-form, line-branch, background, and defect coordinate.

An unqualified phrase “the complete Standard Model symmetry atlas” is insufficient once line, monopole, background, fractionalization, or defect sectors are in scope.

Proof. By [theorem 15.2](#), different global packets are distinct at extended scope. A closure theorem that does not fix or quantify over a distinguishing target coordinate is ambiguous: it can silently import incompatible line, one-form, theta-identification, physical-phase, fractionalization, or response data. The three listed scopes exhaust the honest alternatives of forgetting, fixing, or indexing the global-completion information. $\square$

19.3 Integrated endpoint

Theorem 19.3 (SYM-9 global-form and generalized-symmetry theorem, SYM9-MAIN). *For the frozen Standard Model local gauge algebra, chiral matter and one-Higgs-doublet representation package, integer charge lattice, and anomaly-neutral local architecture:*

1. *the common center kernel is $\mathbb{Z}_6$ and the compatible compact connected quotient family is $\Gamma = 1, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_6$;*
2. *for each Γ and fixed line-spectrum branch $\mathcal{P}_\Gamma$, the electric descent, magnetic/dyonic mutual-locality, screening, one-form, and gauge-topological theta-identification data form a coherent line/topological packet at the declared line and monopole scope;*
3. *the gauge-topological theta torus and the retained mass/Yukawa phase data descend, under admitted anomalous fermion rephasings, to the physical phase packet $\Phi_\Gamma^{\text{phys}}$;*
4. *when mixed background symmetries are admitted, their action and topological response are additionally indexed by the applicable fractionalization or extension datum $\mathbf{F}$;*
5. *different genuine extended, background-action, gauge-topological, or physical-phase packets are distinct completed targets even when the local perturbative sector agrees;*
6. *higher-flavor 2-group structures, with their normalization class $\mathcal{K}_{\mathbf{F},\Gamma}$, and massless noninvertible structures enter only on their explicit target fibers;*
7. *the complete packet is invariant under lawful local and coordinate redescrptions;*
8. *downstream closure must be sharp-local, fixed-target, or explicitly family indexed.*

No clause selects a unique physically realized Γ or infers a unique maximal mutually local line branch from G_Γ alone.

Proof. Clauses 1–7 are the composition of [theorems 6.1, 7.1, 11.2, 12.1, 13.1, 15.2, 16.1, 17.1, 18.2](#) and [19.1](#). Clause 8 is [theorem 19.2](#). The final nonselection statement follows from [theorems 2.4](#) and [5.1](#) and the absence of an additional global-form or line-branch selector premise. $\square$

Proof and certification status

The theorem is a scientific composition of local derivations and primary QFT results. The accompanying archive checks the target registry, encoded center and lattice arithmetic, theorem/source wiring, finite audit rows, and artifact synchronization. Those checks do not prove the field-theory theorem, select a global form or line branch, or establish generalized-symmetry exhaustion beyond the declared targets.

20 Rival analysis and falsification boundary

20.1 Rival classes

A proposed rival is classified only after its target fingerprint is complete. The rival ledger uses the following dispositions:

1. **Equivalent presentation:** same packet under a lawful redescription.
2. **Incomplete claim:** missing global form, operator category, line-spectrum branch, screening, monopole, background, fractionalization, phase quotient, or limit field.

3. **Incompatible:** violates descent, mutual locality on the selected branch, screening, anomaly, theta-identification, physical-phase, background-action, or target equations.
4. **Target update:** changes Γ , line branch, dynamical matter, monopole spectrum, spin structure, backgrounds, fractionalization/extension class, Yukawas, masses, or defect category.
5. **Open same-target burden:** sufficiently specified to challenge the theorem but not resolved by the present proof.
6. **Genuine counterexample:** satisfies every frozen premise and yields non-isomorphic packet data without breaking a cited theorem.

No unregistered proposal is rejected merely because it is unregistered.

20.2 Concrete rival tests

The following examples exercise the theorem boundaries.

Proposal	Disposition	Reason
Treat $SU(3) \times SU(2) \times U(1)$ as globally unique	Incomplete	Local notation omits Γ and line data.
Claim $\Gamma = \mathbb{Z}_6$ is uniquely forced by local matter	Rejected overclaim	Maximal faithful local-matter quotient is not a physical selector among global targets.
Infer a unique maximal magnetic/dyonic line spectrum from G_Γ alone	Incomplete	A line-spectrum branch or discrete theta-like choice is omitted.
Add a Wilson line that fails the quotient congruence	Incompatible	Representation does not descend to G_Γ .
Infer electric one-form symmetry from the center without screening	Incomplete	Dynamical matter endability is omitted.
Keep continuous magnetic $U(1)^{(1)}$ after adding unit monopoles	Incompatible	Bianchi current is explicitly broken.
Change theta period but keep line monodromy fixed	Incompatible	Witten-effect transport fails.
Treat a shift outside the theta lattice as full-theory inequivalence without fixing the matter phases	Incomplete	An anomalous fermion rephasing can transport theta and mass/Yukawa phases together.
Use the same background-field set with a different fractionalization class and call it equivalent	Distinct target	Junction and response data differ.

Copy the source higher-group coefficient unchanged after rescaling the magnetic current	Incompatible	The contragredient background and period-lattice transport is omitted.
Use the $B - L$ response in another zero-form charge normalization without transport	Incomplete	The numerical response belongs to the Hsin–Gomis unit-charge convention.
Use zero-Yukawa higher flavor at physical Yukawas	Target update/controlled breaking	Exact flavor premise is absent.
Use massless non-invertible defects at nonzero fermion mass	Target update	Exact topological conservation is broken.
Infer full Standard Model CP invariance from a CP-fixed gauge-theta class	Rejected overclaim	Yukawa/mixing and other CP-odd sectors remain independent.
Choose another generator of the same Γ	Equivalent presentation	Embedded subgroup and packet are unchanged.
Introduce a disconnected global group	Target update	Outside the compact connected quotient family.

20.3 Faithful counterexample burden

A counterexample to [theorem 19.3](#) must specify:

1. the exact field theory, Γ , operator category, monopole policy, line-spectrum branch, background transformation package, fractionalization/extension class, mass/Yukawa limit, and spin scope;
2. a globally defined line or defect object and its action;
3. its descent, mutual-locality, screening, anomaly, theta-identification, physical-phase, and background-response data;
4. the invariant by which it differs from the canonical packet;
5. why it is not related by a lawful redescription;
6. the first theorem implication it breaks.

A proposal that changes the target is scientifically relevant but does not falsify a fixed-target theorem.

20.4 Adversarial review questions

A substantive adversarial review should ask:

- Is the common-center calculation correct for every field and convention?
- Does the quotient family omit an allowed central subgroup in the declared class?
- Is the selected maximal mutually local line branch explicit, and is any uniqueness claim stronger than the sources support?
- Do all displayed electric lines descend?
- Are globally admissible and unscreened lines distinguished?
- Does the magnetic lattice satisfy DSZ mutual locality on the declared branch?
- Is the one-form group derived after screening and monopole specification?

- Are theta shifts compatible with the selected line branch, and is the gauge-topological torus kept distinct from the full physical phase quotient?
- Are background transformation laws and required fractionalization classes both specified?
- Is every higher-group coefficient quoted with its magnetic-current and two-form-background normalization, including the period lattice?
- Is the $B - L$ response quoted in the charge convention of its source?
- Are gauge-topological CP/T fixed classes kept distinct from full Standard Model CP/T status?
- Are local and extended operator equivalences kept separate?
- Are the higher-group and noninvertible claims confined to their exact targets?
- Does any AASC role label appear as a premise of the QFT proof?
- Does the certification layer claim semantic completeness it has not proved?

20.5 Bounded artifact tests

The accompanying archive includes a bounded mutation suite checking representative corruptions such as a wrong center phase, omitted quotient, invalid descent congruence, broken DSZ basis line, wrong one-form order, wrong theta-lattice determinant, missing line-branch coordinate, missing fractionalization datum, untransported higher-group normalization, missing $B - L$ response convention, CP-scope collapse, mismatched theorem source, or language that promotes the validator into theorem authority. These tests audit the encoded archive. They are not a substitute for the scientific questions above.

Adversarial check

The finite rival registry is complete for its published rows. It is not a proof that every possible line branch, defect, background completion, fractionalization class, or ultraviolet target has been generated. Whole-atlas candidate completeness remains a separate SYM-11 burden.

20.6 Scientific endpoint boundary

The claimed endpoint depends on the field-theory chain through SYM9-MAIN. A wrong center, fixed-branch line, one-form, theta-identification, physical-phase, fractionalization, higher-group, or noninvertible theorem at the declared target is load-bearing. Citation, exposition, or finite artifact defects are separate unless the scientific proof depends on them. Unique selection of Γ , a universal line branch derived from the global group alone, classification of every ultraviolet completion, and SYM-11 exhaustion are outside the stated theorem.

21 Handoff to SYM-10 and SYM-11

21.1 Handoff to SYM-10

SYM-10 studies scale, renormalization-group transport, and emergent symmetry. It receives from SYM-9:

1. the distinction between a fixed global target and a scale-dependent effective description;
2. the requirement that RG matching transport one-form anomalies, line charges, fractionalization/extension data, and higher-group coefficients where the relevant operator category survives;
3. the target-fiber status of massless noninvertible symmetries and zero-Yukawa higher flavor;
4. the rule that integrating out matter can change screening and hence emergent one-form symmetry without retroactively changing the ultraviolet global group or line-branch choice.

An infrared generalized symmetry is therefore recorded with its ultraviolet ancestry, breaking operators, and matching data.

21.2 Handoff to SYM-11

SYM-11 receives a nonoptional global-completion index. Its closure theorem must specify whether it concerns:

- the sharp-local Standard Model, in which global-form discriminants are intentionally outside the operator category;
- one fixed defect-completed target $D_{SM,\Gamma}^{\text{def}}$, with line branch, monopole policy, background-action/fractionalization data, gauge-topological theta torus, physical phase quotient, and all other packet fields declared;
- or an explicitly indexed family of completed targets, with every atlas row and role quantified over the retained Γ , line, monopole, background, and defect coordinates.

The capstone may not merge rows whose local names agree but whose line, one-form, theta-identification, physical-phase, fractionalization, response, or defect status differs.

Final endpoint

The Standard Model local gauge algebra and point-particle representation ledger leave four compact connected global-form targets. A global group fixes electric descent but does not by itself select every maximal mutually local magnetic/dyonic line branch. For each complete target—including Γ , the selected line branch, monopole policy, and applicable background-action/fractionalization data—the line structure, screening, one-form symmetry, gauge-topological theta identifications, physical phase quotient, and admitted higher or noninvertible structures form a coherent packet. Lawful presentation changes preserve the packet; genuine changes in extended, background-response, gauge-topological, or physical-phase content change the target. The result supplies indexed global-completion normal forms and a precise closure boundary, not a unique selection of the physical quotient or line branch.

21.3 Established results

The analysis establishes:

1. the exact common center kernel and quotient family;
2. electric descent and magnetic mutual-locality conditions on the declared line branch;
3. the distinction among admissibility, maximal mutual locality, genuineness, and screening;
4. the global-form and branch dependence of one-form symmetry and gauge-topological theta identifications;
5. the distinction between the gauge-topological theta torus and the full physical phase quotient under anomalous fermion rephasings;
6. the need for explicit fractionalization or extension data in mixed background actions;
7. the electroweak-broken probe-charge interface;
8. physical inequivalence of distinct completed packets;
9. target-specific higher-flavor 2-group and noninvertible defect results;
10. a redescription-stable AASC role map;
11. the SYM-11 fixed-target/family-indexed handoff.

21.4 Scope beyond the theorem

The physical Standard Model global form and line-spectrum branch remain empirical or ultraviolet questions. Disconnected global groups, nonspin refinements, unclassified discrete theta-like branches, complete dynamical monopole spectra, every fractionalization class or generalized symmetry of SMEFT and ultraviolet completions, and quantitative breaking of noninvertible symmetries by physical masses

remain outside the theorem. These are visible open directions, not hidden deficits.

21.5 Conclusion

Local gauge theory does not end with the Lie algebra. Once extended operators are admitted, the global quotient controls which electric probes descend, while a complete line target also fixes which magnetic and dyonic probes form the selected maximal mutually local branch. Matter and monopoles determine which charges are screened, the resulting one-form actions, and the admissible gauge-topological theta identifications. Anomalous fermion rephasings then relate those coordinates to the mass and Yukawa phases in the physical phase quotient. Mixed background symmetries can require further fractionalization or extension data before their action is determinate. The Standard Model therefore comes not as one unqualified global object but as a controlled family of global completions compatible with one local perturbative core.

This family structure is the correct form of rigidity. The packet is not freely editable inside a completed target, yet the local theory does not select one quotient or line branch from the family. Higher-group and noninvertible structures reinforce the same lesson: a symmetry statement is meaningful only with its operator, background, fractionalization, mass, and monopole scope. With those fields frozen, the global and defect sector has a determinate normal form. Without them, the word “symmetry” conceals a target ambiguity.

V

Reference and reproducibility appendices

A Notation, target coordinates, and theorem registry

A.1 Target-coordinate dictionary

A complete SYM-9 statement carries the following coordinates.

Coordinate	Typical values	Function
Local theory D	renormalizable SM; massless QED; massless two-flavor QCD	Fixes fields, interactions, and quantum regime.
Global form Γ	$1, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_6$	Fixes the quotient of the product cover.
Operator category O	local; line; defect/background completed	Fixes which observables can distinguish targets.
Magnetic policy M	no dynamical monopoles; specified monopole lattice; UV completion	Fixes magnetic screening and one-form exactness.
Line polarization P	reference theta class; maximal mutually local dyonic branch; theta monodromy	Fixes the line-spectrum branch not selected by the gauge group alone.
Backgrounds B	none; flavor 1-forms; one-form 2-forms; defect TQFT	Fixes which probes are admitted.
Background action F	not applicable; fractionalization/Postnikov class; response-counterterm class	Fixes how admitted symmetries act on higher-form charged objects.

Matter-phase fiber Y	physical phases; zero-Yukawa; massless QED/QCD	Fixes exactness, controlled breaking, and the matter-phase data entering $\Phi_\Gamma^{\text{phys}}$.
Geometry S	spin Minkowski; specified bundle/topology	Fixes global anomaly and flux quantization scope.

A.2 Symbol dictionary

Symbol	Meaning
$\mathfrak{g}_{\text{SM}}$	Local Standard Model gauge algebra.
G_0	Product cover $\text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_{\tilde{Y}}$.
ξ	Common center generator $(\omega_3, -1, e^{i\pi/3})$.
K_{mat}	Kernel of the product-cover action on local matter; $\mathbb{Z}_6$.
G_Γ	Global quotient G_0/Γ .
$z_{2,e}, z_{3,e}$	Electric center charges modulo 2 and 3.
q	Integer hypercharge $\tilde{Y}$.
$z_{2,m}, z_{3,m}, g$	Magnetic center residues and Abelian magnetic charge.
$\langle \ell, \ell' \rangle$	DSZ pairing of two dyonic line charges.
$\mathcal{S}_\Gamma$	Electric and magnetic screening relation.
$\mathcal{A}_\Gamma^{(1)}$	One-form global symmetry at fixed target.
$\Lambda_{\theta,\Gamma;\mathcal{P}_\Gamma}$	Gauge-topological theta-identification lattice on the selected line branch.
$\Theta_\Gamma^{\text{line}}$	Gauge-topological line torus $\mathbb{R}^3/\Lambda_{\theta,\Gamma;\mathcal{P}_\Gamma}$ with Witten-effect monodromy.
$\Phi_\Gamma^{\text{phys}}$	Physical phase quotient $(\Theta_\Gamma^{\text{line}} \times \mathcal{Y}_{\text{ph}})/\mathcal{R}_{\text{anom}}$.
$\mathcal{P}_\Gamma$	Selected maximal mutually local line-polarization/theta branch.
$\mathcal{F}_\Gamma$	Background-action/fractionalization and response class.
$\mathcal{K}_{F,\Gamma}$	Normalization class of the higher-group current, two-form background, period lattice, and coefficient.
Q_{B-L}^{HG}	Hsin–Gomis charge $\frac{1}{3}B_q - \sum_i L_i$; equivalently $B_{\text{conv}} - \sum_i L_i$, with unit minimal gauge-invariant local charge.
$J_{\text{CK}}^{(2)}$	Magnetic one-form current in the Córdova–Koren source convention.
$J_\Gamma^{(2)}$	Unit-line-charge magnetic current $n \star da_Y/(2\pi)$ for $\Gamma = \mathbb{Z}_n$.
r_Γ	Current-conversion factor defined by $J_\Gamma^{(2)} = r_\Gamma J_{\text{CK}}^{(2)}$.
$\widehat{\kappa}_F^{\text{CK}}$	Córdova–Koren source-coordinate coefficient $-\kappa_F^{\text{CK}}/2$.
$\widehat{\kappa}_{F,\Gamma}$	Transported coefficient $r_\Gamma^{-1}\widehat{\kappa}_F^{\text{CK}}$ with contragredient background and period transport.
$\mathcal{H}_\Gamma^{\text{higher}}$	Higher-group packet including the normalized background law and current/background period convention when its target premises hold.
$\mathcal{N}_\Gamma$	Noninvertible defect packet when its target premises hold.
$\mathbb{G}_\Gamma$	Complete global-form-indexed packet.

A.3 Theorem registry

Table 8: Stable theorem registry and proof authority.

ID	Result	Authority	Status
SYM9-F01	Target and operator-category freeze	local definition	proved
SYM9-F02	Normalization covariance	local proof	proved
SYM9-Q01	Local-to-global underdetermination	local proof plus primary QFT interface	proved
SYM9-Q02	Common-center kernel	local finite calculation	proved
SYM9-Q03	Compact connected global-form family	local group-theoretic proof	proved
SYM9-Q04	Electric-line admissibility	local congruence proof	proved
SYM9-Q05	Screening quotient	local proof plus generalized-symmetry interface	proved
SYM9-Q06	Magnetic/dyonic compatibility	local arithmetic specialized from Tong	proved
SYM9-Q07	Genuine-line reconstruction	local synthesis	proved at fixed line-polarization/charge-lattice scope
SYM9-Q08	One-form reconstruction	local proof plus primary interface	proved at declared monopole scope
SYM9-Q09	Theta-identification and physical phase quotient	primary result with local lattice checks	proved/imported at fixed branch and anomalous-rephasing scope
SYM9-Q10	Global-form separation	local operator-category proof	proved
SYM9-Q11	Higher-group interface with normalization transport	primary QFT import plus local convention transport	proved at zero-Yukawa target with source-normalized coefficient and normalization class K_F, Γ
SYM9-Q12	Noninvertible target-fiber	primary QFT import specialized to target	conditional target-fiber theorem
SYM9-Q13	Global-completion covariance including current/background normalization	local proof	proved with normalization-covariant background laws
SYM9-A01	Global-form role	AASC composition	proved from QFT results
SYM9-A02	Generalized-action role	AASC composition	proved from QFT results
SYM9-A03	Global-completion normal form	local composition	proved at complete target fingerprint

ID	Result	Authority	Status
SYM9-A04	Family-indexed closure handoff	local composition	proved
SYM9-MAIN	Integrated global-form and generalized-symmetry theorem	composition theorem	proved at complete declared target scope

A.4 Dependency order

The stable theorem order is

$$F01 \longrightarrow Q01 \longrightarrow Q02 \longrightarrow Q03 \longrightarrow \begin{matrix} Q04 \rightarrow Q05 \\ Q06 \end{matrix} \longrightarrow Q07 \longrightarrow \begin{matrix} Q08 \\ Q09 \end{matrix} \longrightarrow Q10.$$

The higher-group and noninvertible branches depend on the target freeze and on external primary results:

$$F01 \rightarrow Q11, Q12.$$

All QFT results feed the covariance and structural composition:

$$Q01\text{--}Q12 \rightarrow Q13 \rightarrow A01, A02 \rightarrow A03 \rightarrow A04 \rightarrow \text{MAIN}.$$

No arrow runs from an AASC role theorem back into the physical derivation.

B Center-kernel calculations and normalization audit

B.1 Representation homomorphism

The product center is

$$Z(G_0) = \mathbb{Z}_3 \times \mathbb{Z}_2 \times U(1).$$

Write a general element as

$$z(a, b, \alpha) = (\omega_3^a, (-1)^b, e^{i\alpha}), \quad a \in \mathbb{Z}_3, \quad b \in \mathbb{Z}_2.$$

On a field with triality t , weak parity s , and integer hypercharge q , its action is

$$\chi_{t,s,q}(z) = \omega_3^{at} (-1)^{bs} e^{iq\alpha}.$$

The common matter kernel is the intersection of the kernels of these six characters.

B.2 Solving the kernel without guessing the generator

Triviality on e^c gives

$$e^{6i\alpha} = 1 \implies \alpha = \frac{k\pi}{3} \quad (k \in \mathbb{Z}_6).$$

Triviality on L gives

$$(-1)^b e^{-3i\alpha} = 1 \implies b \equiv k \pmod{2}.$$

Triviality on d^c gives

$$\omega_3^{-a} e^{2i\alpha} = 1 \implies -a + k \equiv 0 \pmod{3},$$

so $a \equiv k \pmod{3}$. Therefore every kernel element is

$$z(k, k, k\pi/3) = \xi^k.$$

The remaining field equations are automatically satisfied; this is checked row by row in [table 9](#).

Table 9: Center-action ledger.

Field	$SU(3)$	$SU(2)$	t	s	$\tilde{Y}$	phase mod 6
Q	3	2	1	1	1	0
u^c	$\bar{3}$	1	-1	0	-4	0
d^c	$\bar{3}$	1	-1	0	2	0
L	1	2	0	1	-3	0
e^c	1	1	0	0	6	0
H	1	2	0	1	3	0

B.3 Subgroups and quotient generators

The four subgroup choices can be displayed as

$$\begin{aligned}
\Gamma_1 &= \{1\}, \\
\Gamma_2 &= \langle \xi^3 \rangle = \langle (1, -1, -1) \rangle, \\
\Gamma_3 &= \langle \xi^2 \rangle = \langle (\omega_3^2, 1, e^{2\pi i/3}) \rangle, \\
\Gamma_6 &= \langle \xi \rangle.
\end{aligned}$$

Replacing ξ^2 by its inverse gives the same embedded $\mathbb{Z}_3$ subgroup. The release registry stores subgroup identity rather than treating a chosen generator tuple as primitive.

B.4 Hypercharge sign and Higgs convention

Some references use the conjugate Higgs doublet and record $\tilde{Y}(H) = -3$ rather than $+3$. Both conventions yield the same center kernel when the Yukawa and field conventions are transported consistently. This manuscript fixes $H : (\mathbf{1}, \mathbf{2})_{+3}$. A source table using the conjugate doublet is not silently merged; its sign convention is recorded in the source ledger.

B.5 Why the local-matter kernel becomes electric one-form symmetry

Before quotienting, K_{mat} is a subgroup of the gauge-group center acting trivially on all dynamical local matter. It therefore acts on Wilson probes but cannot be screened by the frozen matter. After quotienting by Γ , the subgroup identified with gauge identity is removed, leaving K_{mat}/Γ . This is the exact sequence underlying [theorem 9.1](#). The statement is target dependent: adding new dynamical representations can reduce the kernel and the resulting one-form symmetry.

B.6 Encoded audit boundary

The accompanying arithmetic audit recomputes

$$(2t + 3s + q) \bmod 6$$

for every row and checks that the global-form registry contains one subgroup of each order 1, 2, 3, 6. It does not prove that disconnected global groups or nonspin extensions are absent, because those are not in the declared family.

C Line, one-form, and theta reference tables

C.1 Electric and magnetic charge conditions

Table 10: Electric descent and representative magnetic bases.

Γ	Electric descent	Magnetic condition	Basis 1	Basis 2
1	none	$z_{2,m} = z_{3,m} = 0, g \in \mathbb{Z}$	$(0, 0; 1)$	–
$\mathbb{Z}_2$	$q \equiv z_{2,e} \pmod{2}$	$z_{3,m} = 0, g \equiv z_{2,m}/2 \pmod{1}$	$(1, 0; 1/2)$	–
$\mathbb{Z}_3$	$q \equiv z_{3,e} \pmod{3}$	$z_{2,m} = 0, g \equiv z_{3,m}/3 \pmod{1}$	$(0, 1; 1/3)$	–
$\mathbb{Z}_6$	$q \equiv 3z_{2,e} - 2z_{3,e} \pmod{6}$	$6g \equiv 3z_{2,m} + 2z_{3,m} \pmod{6}$	$(1, 0; 1/2)$	$(0, 1; 1/3)$

The table stores center-charge conditions, not full highest-weight data. Within each center class there can be many representations. The center class is sufficient for quotient descent and DSZ congruence but not for every dynamical property of the line.

C.2 Derivation of the electric conditions

For $\Gamma = \mathbb{Z}_n = \langle \xi^{6/n} \rangle$, the phase of an electric line under the generator is

$$\exp \left[\frac{2\pi i}{n} \frac{3z_{2,e} + 2z_{3,e} + q}{1} \right].$$

Triviality gives

$$3z_{2,e} + 2z_{3,e} + q \equiv 0 \pmod{n}.$$

Reducing this equation for $n = 2, 3, 6$ yields the displayed rules.

C.3 Representative DSZ checks

For the $\mathbb{Z}_2$ magnetic basis $(z_{2,m}, z_{3,m}; g) = (1, 0; 1/2)$ and an allowed electric line $q = z_{2,e} + 2k$,

$$\begin{aligned} 3z_{2,e} - 6q \left(\frac{1}{2} \right) &= 3z_{2,e} - 3(z_{2,e} + 2k) \\ &= -6k \in 6\mathbb{Z}. \end{aligned}$$

For the $\mathbb{Z}_3$ basis $(0, 1; 1/3)$ and $q = z_{3,e} + 3k$,

$$2z_{3,e} - 6q \left(\frac{1}{3} \right) = 2z_{3,e} - 2(z_{3,e} + 3k) = -6k.$$

For $\mathbb{Z}_6$, either basis satisfies the pairing with $q = 3z_{2,e} - 2z_{3,e} + 6k$. These computations demonstrate the compatibility of the displayed generators. The full mutually local dyonic lattice includes their integer combinations and theta-shifted images.

C.4 One-form and electroweak comparison

Table 11: Global-form packet summary in the no-dynamical-monopole target.

Γ	n	Electric 1-form	Magnetic 1-form	$\tilde{\theta}$ period	$ Q _{\min}$	$ G _{\min}$
1	1	$\mathbb{Z}_6$	U(1)	2π	1/6	6
$\mathbb{Z}_2$	2	$\mathbb{Z}_3$	U(1)	8π	1/3	3
$\mathbb{Z}_3$	3	$\mathbb{Z}_2$	U(1)	18π	1/6	2
$\mathbb{Z}_6$	6	1	U(1)	72π	1/3	1

The electric one-form factor is the residual local-matter kernel. The magnetic factor shown is conditional on M_0 . The electromagnetic minimum charges are line-probe minima after one-doublet fixation and should not be read as observed new particles.

C.5 Gauge-topological theta-identification lattice

Table 12: Basis vectors for $\Lambda_{\theta,\Gamma}$ in the coordinate order $(\theta_2, \theta_3, \tilde{\theta})$.

Γ	Basis index	Vector
1	1	$(2\pi, 0\pi, 0\pi)$
1	2	$(0\pi, 2\pi, 0\pi)$
1	3	$(0\pi, 0\pi, 2\pi)$
$\mathbb{Z}_2$	1	$(2\pi, 0\pi, 4\pi)$
$\mathbb{Z}_2$	2	$(0\pi, 2\pi, 0\pi)$
$\mathbb{Z}_2$	3	$(0\pi, 0\pi, 8\pi)$
$\mathbb{Z}_3$	1	$(2\pi, 0\pi, 0\pi)$
$\mathbb{Z}_3$	2	$(0\pi, 2\pi, 6\pi)$
$\mathbb{Z}_3$	3	$(0\pi, 0\pi, 18\pi)$
$\mathbb{Z}_6$	1	$(2\pi, 0\pi, 36\pi)$
$\mathbb{Z}_6$	2	$(0\pi, 2\pi, 48\pi)$
$\mathbb{Z}_6$	3	$(0\pi, 0\pi, 72\pi)$

Dividing every basis vector by 2π gives integral/rational matrices with determinants

$$1, \quad 4, \quad 9, \quad 36,$$

which equal n^2 . This is the finite arithmetic relation recorded in the accompanying audit files.

C.6 CP/T-fixed classes of the gauge-topological theta sector

For convenience, representatives of the fixed classes under $\vartheta \mapsto -\vartheta$ are:

Γ	Representatives
1	$(\theta_2, \theta_3, \tilde{\theta})$ with each coordinate 0 or π .
$\mathbb{Z}_2$	$(0, 0/\pi, 0/4\pi)$ and $(\pi, 0/\pi, 2\pi/6\pi)$, with the middle entry independently 0 or π .
$\mathbb{Z}_3$	$(0/\pi, 0, 0/9\pi)$ and $(0/\pi, \pi, 3\pi/12\pi)$.
$\mathbb{Z}_6$	$(0, 0, 0/36\pi)$; $(0, \pi, 12\pi/48\pi)$; $(\pi, 0, 18\pi/54\pi)$; $(\pi, \pi, 30\pi/66\pi)$.

The slash notation separates two representatives, not a quotient operation. These are fixed classes of the gauge-topological theta packet only; they do not establish CP or time-reversal invariance of the full Standard Model Yukawa/mixing sector.

C.7 Anomaly-invariant weak/hypercharge combination and phase quotient

An anomalous baryon or lepton rotation shifts $(\theta_2, \tilde{\theta})$ along the vector $(-1, 18)$. The invariant linear functional is

$$\tilde{\theta} + 18\theta_2.$$

After electroweak fixation it projects to $4\theta_{\text{em}}$. More generally, if $\mathcal{Y}_{\text{ph}}$ denotes the mass/Yukawa phase data and $\mathcal{R}_{\text{anom}}$ the admitted anomalous rephasings, the full physical phase record is

$$\Phi_{\Gamma}^{\text{phys}} = (\Theta_{\Gamma}^{\text{line}} \times \mathcal{Y}_{\text{ph}}) / \mathcal{R}_{\text{anom}}.$$

The gauge-topological torus $\Theta_{\Gamma}^{\text{line}}$ controls line monodromy at fixed matter-phase convention; $\Phi_{\Gamma}^{\text{phys}}$ controls physical equivalence after the matter phases are transported.

D Atlas projection, imports, and source provenance

D.1 Atlas projection

The full arc atlas has thirty fields. The table below is the publication projection containing identity, domain, quotient-action status, role, and theorem owner. The machine-readable CSV and JSON retain the same row identities.

Table 14: SYM-9 atlas projection.

Record ID	Name	Domain	Role	Owner
SYM-09-GFORM-LOCAL-UNDERDET-v1	Local/global underdetermination	local and line categories	target datum	Q01
SYM-09-GFORM-KERNEL-Z6-v1	Common matter kernel Z6	local representation package	derived witness	Q02
SYM-09-GFORM-G1-v1	Global form $\Gamma=1$	line-completed target	target datum	Q03
SYM-09-GFORM-G2-v1	Global form $\Gamma=Z2$	line-completed target	target datum	Q03
SYM-09-GFORM-G3-v1	Global form $\Gamma=Z3$	line-completed target	target datum	Q03
SYM-09-GFORM-G6-v1	Global form $\Gamma=Z6$	line-completed target	target datum	Q03
SYM-09-LINE-ELECTRIC-v1	Electric line descent	line category	physical/witness	Q04
SYM-09-LINE-SCREEN-v1	Screening quotient	line category and matter	physical/witness	Q05
SYM-09-LINE-MAG-v1	Magnetic and dyonic lattice	fixed- Γ fixed-polarization	physical/witness	Q06
SYM-09-1FORM-ELECTRIC-v1	Electric one-form symmetry	line category	physical action	Q08
SYM-09-1FORM-MAG-v1	Magnetic one-form symmetry	M0 target	physical action	Q08
SYM-09-THETA-LATTICE-v1	Gauge-topological theta-identification torus	fixed line branch and matter-phase convention	target/witness	Q09
SYM-09-PHASE-QUOTIENT-v1	Physical theta/matter phase quotient	fixed global target with mass/Yukawa phase data	physical invariant/witness	Q09/Q13
SYM-09-EW-LINES-v1	EW-broken line projection	Higgs phase line target	physical/witness	EW interface
SYM-09-HIGHER-FLAVOR-v1	Higher flavor 2-group with normalization-covariant extension datum	zero-Yukawa background target with current/background/period normalization class	physical action/effective route	Q11
SYM-09-NONINV-QED-v1	Massless QED noninvertible defects	massless defect target	physical action/target fiber	Q12
SYM-09-NONINV-QCD-v1	Massless QCD noninvertible defects	massless IR defect target	physical action/matching witness	Q12
SYM-09-COVARIANCE-v1	Global-completion covariance	fixed completed target	skin rule	Q13
SYM-09-HANDOFF-v1	Family-indexed closure scope	SYM11 interface	target rule	A04
SYM-09-LINE-POLARIZATION-v1	Line-polarization/theta-branch datum	fixed global-form line target	target datum	Q07
SYM-09-BG-FRACTIONALIZATION-v1	Background-action/fractionalization class	mixed zero-/one-form background target	target datum/witness	Q10/A03

Table 15: Frozen target and target-fiber registry. The line-polarization and background-action columns are target coordinates fixed by the theorem statements.

Target ID	Global form	Operator	Magnetic	Line branch P	Backgrounds B	Action/fractionalization F	Yukawa/mass scope
SYM9-T-LOC	UNFIXED_FAMILY	O_{loc}	–	NA	none	NA	generic renormalizable SM with retained mass/Yukawa phase data
SYM9-T-G1	$\Gamma = 1$	O_{line}	M_0	Tong reference branch at $\theta=0$ plus lattice monodromy	none	NA	generic renormalizable SM with retained mass/Yukawa phase data
SYM9-T-G2	$\Gamma = \mathbb{Z}_2$	O_{line}	M_0	Tong reference branch at $\theta=0$ plus lattice monodromy	none	NA	generic renormalizable SM with retained mass/Yukawa phase data
SYM9-T-G3	$\Gamma = \mathbb{Z}_3$	O_{line}	M_0	Tong reference branch at $\theta=0$ plus lattice monodromy	none	NA	generic renormalizable SM with retained mass/Yukawa phase data
SYM9-T-G6	$\Gamma = \mathbb{Z}_6$	O_{line}	M_0	Tong reference branch at $\theta=0$ plus lattice monodromy	none	NA	generic renormalizable SM with retained mass/Yukawa phase data
SYM9-T-BLBG	Γ -indexed	O_{def}	M_0	inherited fixed-Gamma reference branch	B-L and magnetic one-form	k in $H^2(\text{BU}(1)_{[B-L]}, \mathbb{Z}_6/\text{Gamma})$; $Q_{B-L} \text{HG} = (1/3) \sum_i B_{q-\text{sum}_i} L_i$ with unit minimal gauge-invariant local charge	generic renormalizable SM with retained mass/Yukawa phase data
SYM9-T-HF	Γ -indexed	O_{def}	M_0 or declared UV	inherited fixed-Gamma branch	flavor and magnetic one-form	higher-group normalization class: magnetic current, two-form background, period lattice, and coefficient	zero-Yukawa limit
SYM9-T-NIQED	electromagnetic line target	O_{def}	probe 't Hooft lines	fixed probe-line branch of cited defect construction	defect TQFT	defect action class fixed; fractionalization NA	massless QED
SYM9-T-NIQCD	IR color-electromagnetic target	O_{def}	probe 't Hooft lines	fixed probe-line branch of cited defect construction	defect TQFT and pion EFT	defect action/matching class fixed; fractionalization NA	massless first-generation QCD below EW scale

D.2 Source authority

Table 16: Source and proof-authority projection. Verification status is retained in the machine-readable ledger.

Source ID	Key	Authority	Use
SRC-P01	Tong2017	primary	Line operators, global forms, theta lattice, EW broken charges
SRC-P02	AharonySeibergTachikawa2013	primary	local/global and line-operator distinction
SRC-P03	GaiottoKapustinSeibergWillett2015	primary	generalized global symmetry framework
SRC-P04	CordovaDumitrescuIntriligator2019	primary	continuous 2-group background law
SRC-P05	BeniniCordovaHsin2019	primary	2-group anomaly framework
SRC-P06	CordovaKoren2023	primary	higher flavor in zero-Yukawa SM and source current/background coefficient convention
SRC-P07	ChoiLamShao2022	primary	massless QED/QCD noninvertible defects
SRC-P08	HsinGomis2024	primary	one-form group, B-L charge convention, fractionalization, and global-form-sensitive response
SRC-P09	DavighiGripaiosLohitsiri2020	primary	global anomaly scope
SRC-P10	AlonsoEtAl2026	primary	Standard Model dyonic lattices, theta domains, CP points, and anomalous B+L reduction to physical theta space
SRC-A01	SYM1	AASC import	local gauge architecture and deferral of global form
SRC-A02	SYM3	AASC import	anomaly typing
SRC-A03	SYM5	AASC import	flavor limit and spurion interface
SRC-A04	SYM8	AASC import	EW residual generator and point spectrum
SRC-A05	AASCTransitionGroupoid	AASC import	local transition groupoid
SRC-A06	AASCCConnection	AASC import	transport geometry
SRC-A07	AASCCurvature	AASC import	loop/holonomy obstruction
SRC-A08	AASCLieType	AASC import	local Lie specialization boundary
SRC-A09	AASCPProductGauge	AASC import	product sector compatibility
SRC-A10	AASCREpAdmissibility	AASC import	representation response
SRC-A11	AASCREpAlgebra	AASC import	algebraic representation coordinates
SRC-A12	AASCAbelianCharge	AASC import	normalization and charge-sensitive quotient
SRC-A13	AASCChargeLattice	AASC import	SM charge lattice
SRC-A14	AASCAnomalyObstruction	AASC import	local/global anomaly discipline
SRC-A15	AASCAnomalyAlgebra	AASC import	trace/congruence anomaly coordinates
SRC-A16	AASCHolonomy	AASC import	discrete holonomy typing
SRC-A17	AASCCPPhases	AASC import	phase/holonomy representative discipline
SRC-A18	AASCStrongCP	AASC import	theta quotient and topological status

D.3 Import exclusions

The import registry records not only what is imported but what is not licensed. Representative examples are:

- SYM-1 licenses the local architecture/operation split, not a global quotient.
- SYM-3 licenses anomaly typing, not quotient selection.
- SYM-5 licenses the zero-Yukawa flavor target, not the higher-group conclusion.
- SYM-8 licenses $Q = T_3 + Y$ and the local electroweak spectrum, not the faithful residual global group.
- charge-lattice sources license the representation and normalization data, not the physical realization of one Γ .
- holonomy and phase sources license representative/invariant discipline, not the Standard Model theta lattice.

These exclusions are part of the dependency firewall and are validated as populated fields in the project data.

D.4 Primary-source priority

The line, one-form, theta, 2-group, and noninvertible claims are anchored in primary technical literature. The AASC corpus is used to preserve target and role discipline. This division ensures that a reader can evaluate the QFT result without accepting the AASC interpretation.

E Exact nonclaims and theorem boundaries

Table 17: Exact nonclaim registry.

ID	Blocked inference
NC01	No unique physical selection of Gamma
NC02	No classification of disconnected or noncompact global groups
NC03	No complete nonspin or spin-charge refinement
NC04	No claim every allowed line is a dynamical particle
NC05	No dynamical monopole prediction
NC06	No generalized symmetry inferred without screening and monopole scope
NC07	No exact $U(3)^5$ higher symmetry at physical Yukawas
NC08	No exact massless noninvertible symmetry at nonzero fermion mass
NC09	No numerical theta-angle determination
NC10	No full UV completion or GUT selection
NC11	No whole-SM symmetry exhaustion
NC12	No machine validator as semantic proof
NC13	No unique maximal dyonic line branch inferred from the gauge group and DSZ condition alone
NC14	No background symmetry action inferred from the background-field set when a fractionalization class is required
NC15	No full Standard Model CP or time-reversal invariance inferred from a gauge-topological theta fixed class
NC16	No identification of the gauge-topological theta torus with the complete physical phase space before anomalous rephasings and mass/Yukawa phases are included

ID	Blocked inference
NC17	No bare higher-group coefficient is treated as invariant under magnetic-current and two-form-background rescaling
NC18	No Hsin-Gomis B-L response coefficient is used outside the charge convention $Q_{B-L}^{HG} = (1/3) B_q - \sum_i L_i$ without transport

E.1 Selection boundary

The maximal quotient $G_{\mathbb{Z}_6}$ is distinguished because it acts faithfully on the frozen local matter package. That mathematical fact does not choose the physical global form. The smaller quotients have additional heavy-probe sectors and nontrivial electric one-form symmetry. A physical selection theorem would need new evidence or ultraviolet premises.

E.2 Completeness boundary

The four-form theorem is complete only within the compact connected central-quotient family of the fixed product cover. It does not classify disconnected extensions, alternative Abelian embeddings, nonspin refinements, or all possible global forms after new matter is added. The line normal form is a charge-level result inside an explicitly fixed maximal mutually local line-polarization/ θ branch; the gauge group and DSZ pairing alone are not asserted to select that branch. The gauge-topological torus $\Theta_{\Gamma}^{\text{line}}$ is not identified with the complete physical phase packet $\Phi_{\Gamma}^{\text{phys}}$, which also carries the mass/Yukawa phases and anomalous-rephasing quotient. The theorem does not reconstruct every line junction or dynamical tension.

E.3 Generalized-symmetry boundary

The one-form theorem is exact at its declared dynamical matter, line-polarization, and monopole scope. A background-field list does not replace a required fractionalization/background-action class. A higher-group coefficient may not be compared across magnetic-current conventions without the contragredient two-form-background and period-lattice transport. The higher-group theorem is exact in the zero-Yukawa background target. The noninvertible theorem is exact in the specified massless defect targets. None of these target qualifiers is optional prose.

E.4 Certification boundary

The accompanying finite data and scripts provide evidence that the published target rows, formulas, and artifacts agree. They do not discover unencoded physical structures and do not prove semantic exhaustion. A new fully scoped same-target proposal must be assessed against the mathematics rather than rejected because it is absent from the registry.

F Reproducibility and certification boundary

F.1 Project layout

The publication archive contains:

- the LaTeX manuscript, style file, bibliography, and generated PDF;
- machine-readable target, theorem, center, global-form, line, one-form, θ , phase-boundary, source, atlas, rival, nonclaim, and adversarial-test registries;
- scripts that regenerate publication tables, validate encoded identities, and execute bounded adversarial mutations;
- JSON audit reports, PDF preflight information, and a SHA-256 manifest.

F.2 Build procedure

A clean build uses

```
make tables
make validate
make adversarial
make pdf
```

The LaTeX build uses `latexmk` and `biber`. The tables are generated from the canonical CSV registries before compilation.

F.3 Encoded checks

The validator recomputes or checks:

- the center phase $2t + 3s + q$ modulo six for every field;
- the exact global-form family and quotient orders;
- the $\mathbb{Z}_6$ electric descent of every local matter field;
- representative DSZ pairings for the displayed magnetic bases;
- the theta-lattice determinants 1, 4, 9, 36;
- stable theorem IDs and source keys;
- atlas CSV/JSON synchronization;
- generated table presence and manuscript section identity;
- absence of unfinished editorial markers and forbidden certificate overclaims.

F.4 What a passing report means

A passing report means that the encoded finite release is internally synchronized and that the tested arithmetic agrees with the manuscript. The certificate layer does not substitute for the scientific proof. It does not mean that a Python program has constructed the full Standard Model line category, proved the higher-group theorem, or exhausted all generalized symmetries. Those claims retain their mathematical and primary-source authority.

F.5 Manifest discipline

The final manifest is computed after the PDF and audit reports are frozen. Verifying it on a pristine extraction establishes file identity. Rerunning the build can update timestamp-bearing reports; a post-run difference in those generated artifacts is not evidence that the pristine archive was corrupt.

F.6 Validation summary

Encoded validation report

The publication audit reports **195/195** checks passed. Its scope is target-field presence, registry identity, encoded center/line/theta arithmetic, the theta/phase and higher-group-normalization boundaries, the declared $B - L$ convention, provenance, table synchronization, and artifact structure. It is not a semantic physics-exhaustion proof and does not select a line polarization.

F.7 Adversarial-test summary

Bounded adversarial artifact-test report

The bounded mutation suite reports **24/24** audits passed. The suite tests the finite encoded publication boundary, including line polarization, fractionalization, CP/T scope, the theta/physical-phase distinction, higher-group normalization transport, the $B - L$ charge convention, and current-source coverage. It does not replace scientific review.

G Bounded adversarial artifact tests

The artifact suite tests whether the published archive detects corruption of the finite claims it explicitly undertakes to protect. It covers center phases, quotient and line-charge records, one-form and theta arithmetic, theorem and source identities, target fields, and the boundary between scientific proof and certification evidence.

G.1 Scope and interpretation

Each mutation changes one encoded assertion while leaving the remaining archive intact. A passing result means that the corresponding inconsistency is rejected by the declared finite checks. It does not show that the archive can invent unknown global forms, derive an anomaly from first principles, select the physical global group, or establish semantic completeness of the line and defect categories.

G.2 Scientific review remains separate

A scientific referee may challenge any line-operator calculation, source specialization, target choice, or conceptual implication independently of the artifact results. Passing the mutation suite does not answer such an objection. The falsification boundary in [section 15–theorem 19.3](#) and the rival protocol remain the scientific gate.

G.3 Registered artifact tests

Table 18: Bounded adversarial artifact-test registry.

ID	Mutation	Description	Expected detection
H01	center_phase_corruption	change Q hypercharge by one	center arithmetic fails
H02	missing_global_form	remove Z3 row	family completeness check fails
H03	wrong_z2_descent	replace $q=z2 \bmod 2$ with $q=0$	electric rule check fails
H04	wrong_z6_descent	flip z3 coefficient	matter descent check fails
H05	bad_z2_magnetic_basis	set $g=1/3$	DSZ check fails
H06	bad_z3_magnetic_basis	set $g=1/2$	DSZ check fails
H07	bad_one_form_order	set $\Gamma=Z2$ electric factor Z2	one-form quotient check fails
H08	bad_theta_period	set Z3 Abelian period 12π	theta determinant check fails
H09	bad_theta_coupling	remove Z6 compensating shift	theta basis check fails
H10	theorem_id_drift	rename Q09	stable theorem set fails
H11	source_authority_drift	mark primary line theorem as local validator proof	authority check fails
H12	unique_gamma_overclaim	insert unique Z6 selection phrase	claim-boundary scan fails
H13	validator_exhaustion_overclaim	insert semantic exhaustion claim	certificate-boundary scan fails
H14	target_fiber_collapse	mark higher flavor generic exact	target-scope check fails
H15	massless_scope_colapse	mark noninvertible generic massive exact	target-scope check fails
H16	artifact_table_drift	alter generated global-form count	table regeneration comparison fails
H17	missing_line_polarization	erase line-branch field from a line-completed target	target fingerprint check fails
H18	missing_fractionalization_class	erase B-L fractionalization class while retaining mixed backgrounds	background-action target check fails
H19	full_cp_overclaim	replace gauge-topological fixed-locus language by full Standard Model CP invariance	scope-language check fails

ID	Mutation	Description	Expected detection
H20	physical_theta_space_collapse	rename gauge-topological line torus as complete physical theta space	theta/phase quotient boundary check fails
H21	outside_lattice_full_inequivalence	claim every outside-lattice shift is full-theory inequivalence without transporting matter phases	anomalous-rephasing boundary check fails
H22	untransported_higher_group_coefficient	copy source kappa-hat unchanged after quotient magnetic-current rescaling	normalization-class boundary check fails
H23	missing_bl_charge_convention	retain fractional response while removing $Q_{B-L}^{HG} = (1/3)B_q - \sum_i L_i$ convention	B-L source-normalization check fails
H24	missing_current_dyonic_source	remove Alonso et al. 2026 dyonic-lattice/theta-space source and positioning	current-primary-source check fails

References

- [1] O. Aharony, N. Seiberg, and Y. Tachikawa. “Reading Between the Lines of Four-Dimensional Gauge Theories”. In: *Journal of High Energy Physics* 2013.8 (2013), p. 115. doi: [10.1007/JHEP08\(2013\)115](https://doi.org/10.1007/JHEP08(2013)115). arXiv: [1305.0318](https://arxiv.org/abs/1305.0318) [hep-th].
- [2] D. Tong. “Line Operators in the Standard Model”. In: *Journal of High Energy Physics* 2017.7 (2017), p. 104. doi: [10.1007/JHEP07\(2017\)104](https://doi.org/10.1007/JHEP07(2017)104). arXiv: [1705.01853](https://arxiv.org/abs/1705.01853) [hep-th].
- [3] D. Gaiotto, A. Kapustin, N. Seiberg, and B. Willett. “Generalized Global Symmetries”. In: *Journal of High Energy Physics* 2015.2 (2015), p. 172. doi: [10.1007/JHEP02\(2015\)172](https://doi.org/10.1007/JHEP02(2015)172). arXiv: [1412.5148](https://arxiv.org/abs/1412.5148) [hep-th].
- [4] A. J. Maley. *The Symmetry Status Calculus: Constraint, Redundancy, Automorphism, and Standing*. SYM-0, AASC corpus manuscript. 2026.
- [5] A. J. Maley. *Gauge Architecture and Gauge Skin in the Standard Model*. SYM-1, AASC corpus manuscript. 2026.
- [6] A. J. Maley. *Anomalies as Structural Obstructions*. SYM-3, AASC corpus manuscript. 2026.
- [7] A. J. Maley. *Flavor Symmetry, Basis Skin, Spurion, and Invariant Normal Form*. SYM-5, AASC corpus manuscript. 2026.
- [8] A. J. Maley. *Accidental Symmetry by Operator Dimension*. SYM-7, AASC corpus manuscript. 2026.
- [9] A. J. Maley. *Electroweak Vacuum Fixation and Residual-Spectrum Rigidity*. SYM-8, publication edition v1.3.0, AASC corpus manuscript. 2026.
- [10] A. J. Maley. *Symmetry as Constraint, Redescription, and Physical Action: A Detailed Research Arc for Classifying the Symmetries of the Standard Model within the Carrier/Skin Program*. Standard Model Symmetry Arc plan, hardened 11 July 2026 edition. 2026.
- [11] Y. Choi, H. T. Lam, and S.-H. Shao. “Noninvertible Global Symmetries in the Standard Model”. In: *Physical Review Letters* 129 (2022), p. 161601. doi: [10.1103/PhysRevLett.129.161601](https://doi.org/10.1103/PhysRevLett.129.161601). arXiv: [2205.05086](https://arxiv.org/abs/2205.05086) [hep-th].
- [12] P.-S. Hsin and J. Gomis. “Detecting Standard Model Gauge Group from Generalized Fractional Quantum Hall Effect”. In: (2024). arXiv: [2411.18160](https://arxiv.org/abs/2411.18160) [hep-th].
- [13] R. Alonso, F. Chadha-Day, D. Dimakou, Y. Ha, and V. V. Khoze. “Dyonic Lattices, θ -Angles and Axions in the Standard Model”. In: (2026). arXiv: [2606.20319](https://arxiv.org/abs/2606.20319) [hep-ph].
- [14] C. Córdova, T. T. Dumitrescu, and K. Intriligator. “Exploring 2-Group Global Symmetries”. In: *Journal of High Energy Physics* 2019.2 (2019), p. 184. doi: [10.1007/JHEP02\(2019\)184](https://doi.org/10.1007/JHEP02(2019)184). arXiv: [1802.04790](https://arxiv.org/abs/1802.04790) [hep-th].
- [15] C. Córdova and S. Koren. “Higher Flavor Symmetries in the Standard Model”. In: *Annalen der Physik* 535.8 (2023), p. 2300031. doi: [10.1002/andp.202300031](https://doi.org/10.1002/andp.202300031). arXiv: [2212.13193](https://arxiv.org/abs/2212.13193) [hep-ph].
- [16] A. J. Maley. *The Gauge Transition Groupoid Theorem: Boundary-Fixed Local Presentation Rigidity under Realized Physical Admissibility*. AASC corpus manuscript. 2026.
- [17] A. J. Maley. *Connection as Standing-Preserving Transport Geometry: Minimal Transport-Comparison Structure for Physical Gauge Realization*. AASC corpus manuscript. 2026.
- [18] A. J. Maley. *Curvature as Transport-Holonomy Obstruction: Boundary-Detectable Loop Residue in Physical Gauge Realization*. AASC corpus manuscript. 2026.
- [19] A. J. Maley. *Lie-Type Gauge Realization Admissibility: When Boundary-Fixed Gauge Transition, Transport, and Loop-Residue Structures Admit Smooth Infinitesimal Specialization*. AASC corpus manuscript. 2026.

- [20] A. J. Maley. *Product Gauge-Sector Compatibility: When Multiple Lie-Type Gauge Ledgers Can Coexist in One Standing Domain*. AASC corpus manuscript. 2026.
- [21] A. J. Maley. *Representation Admissibility and Charge-Witness Rigidity: Standing-Preserving Representation Response in Physical Gauge Realization*. AASC corpus manuscript. 2026.
- [22] A. J. Maley. *Representation Algebra Specialization: When Representation-Response Carriers Admit Algebraic Representation Spaces*. AASC corpus manuscript. 2026.
- [23] A. J. Maley. *Abelian Charge-Witness Normalization: Charge Lattice Readiness, Normalization Skin, and Abelian Factor Rigidity*. AASC corpus manuscript. 2026.
- [24] A. J. Maley. *Rigidity of the Standard-Model Charge Lattice: A Fixed-Scope Reconstruction Theorem for Charge Quantization*. AASC corpus manuscript. 2026.
- [25] A. J. Maley. *Anomaly as Standing-Failure Obstruction: Cancellation as an Admissibility Requirement for Representation-Bearing Gauge Realization*. AASC corpus manuscript. 2026.
- [26] A. J. Maley. *Anomaly Ledger Algebraization: When Standing-Obstruction Neutrality Becomes Trace, Sum, Probe-Induced Charge, and Index Cancellation*. AASC corpus manuscript. 2026.
- [27] J. Davighi, B. Gripaios, and N. Lohitsiri. “Global Anomalies in the Standard Model(s) and Beyond”. In: *Journal of High Energy Physics* 2020.7 (2020), p. 232. doi: [10.1007/JHEP07\(2020\)232](https://doi.org/10.1007/JHEP07(2020)232). arXiv: [1910.11277](https://arxiv.org/abs/1910.11277) [hep-th].
- [28] A. J. Maley. *Discrete Holonomy Exactness*. AASC corpus manuscript. 2026.
- [29] A. J. Maley. *CP-like Phases as Admissible Holonomy Classes*. AASC corpus manuscript. 2026.
- [30] A. J. Maley. *The Strong-CP Boundary Endpoint: An AASC Constraint Theorem for CP-Odd QCD Topological Standing*. AASC corpus manuscript. 2026.
- [31] F. Benini, C. Córdova, and P.-S. Hsin. “On 2-Group Global Symmetries and Their Anomalies”. In: *Journal of High Energy Physics* 2019.3 (2019), p. 118. doi: [10.1007/JHEP03\(2019\)118](https://doi.org/10.1007/JHEP03(2019)118). arXiv: [1803.09336](https://arxiv.org/abs/1803.09336) [hep-th].
- [32] E. Witten. “Dyons of Charge $e\theta/2\pi$ ”. In: *Physics Letters B* 86 (1979), pp. 283–287. doi: [10.1016/0370-2693\(79\)90838-4](https://doi.org/10.1016/0370-2693(79)90838-4).
- [33] A. J. Maley. *The Discrete Symmetry Ledger of the Standard Model*. SYM-4, AASC corpus manuscript. 2026.